



Teaching Assistants

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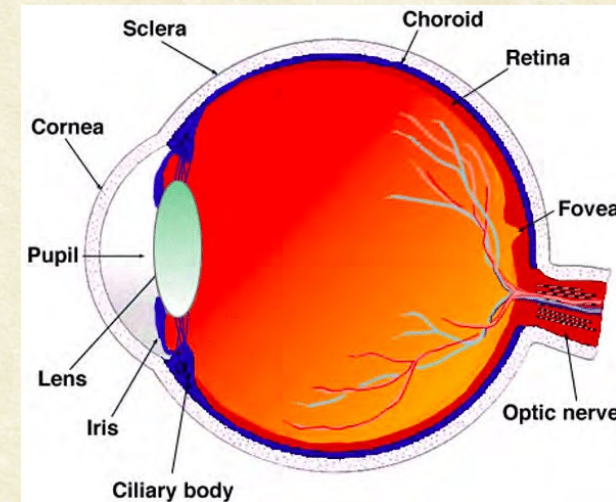
Topics for Today

- I. Human Visual System
- II. Image Acquisition: Digital Image Sensors
- III. Image Signals: Representation and Properties
- IV. Manipulating Pixels



CS7.404: Digital Image Processing

Monsoon 2025: I: Human Visual System

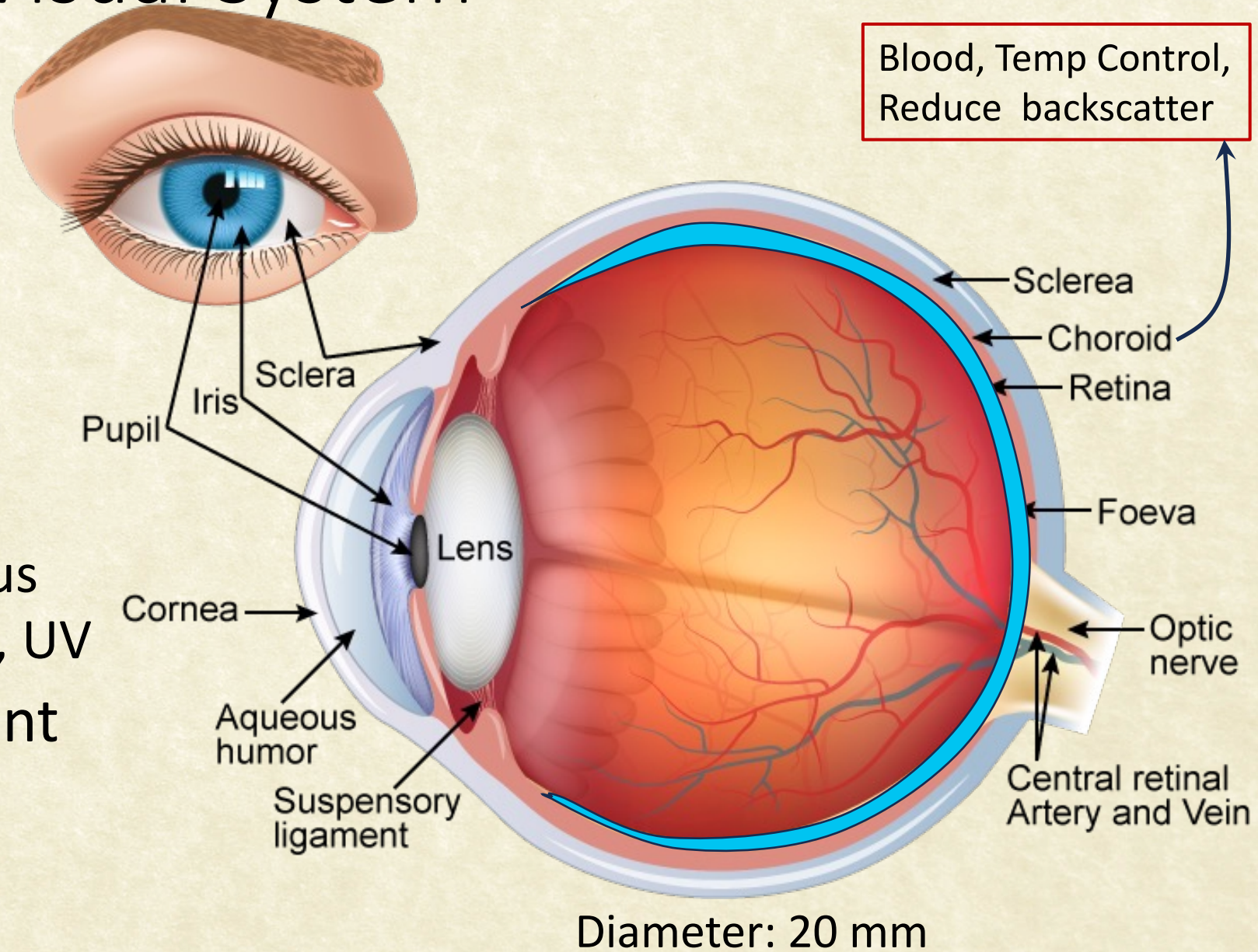


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The Human Visual System

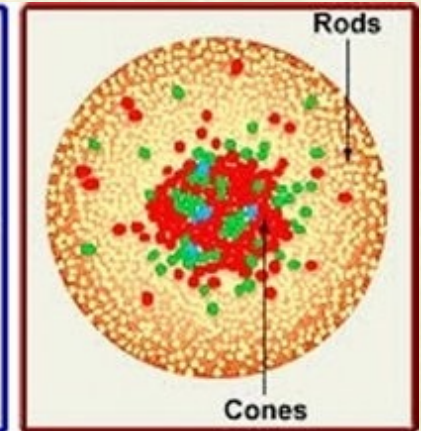
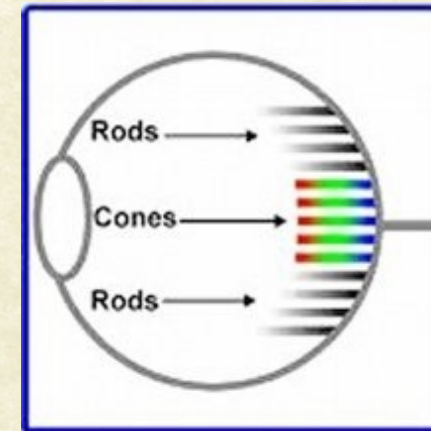
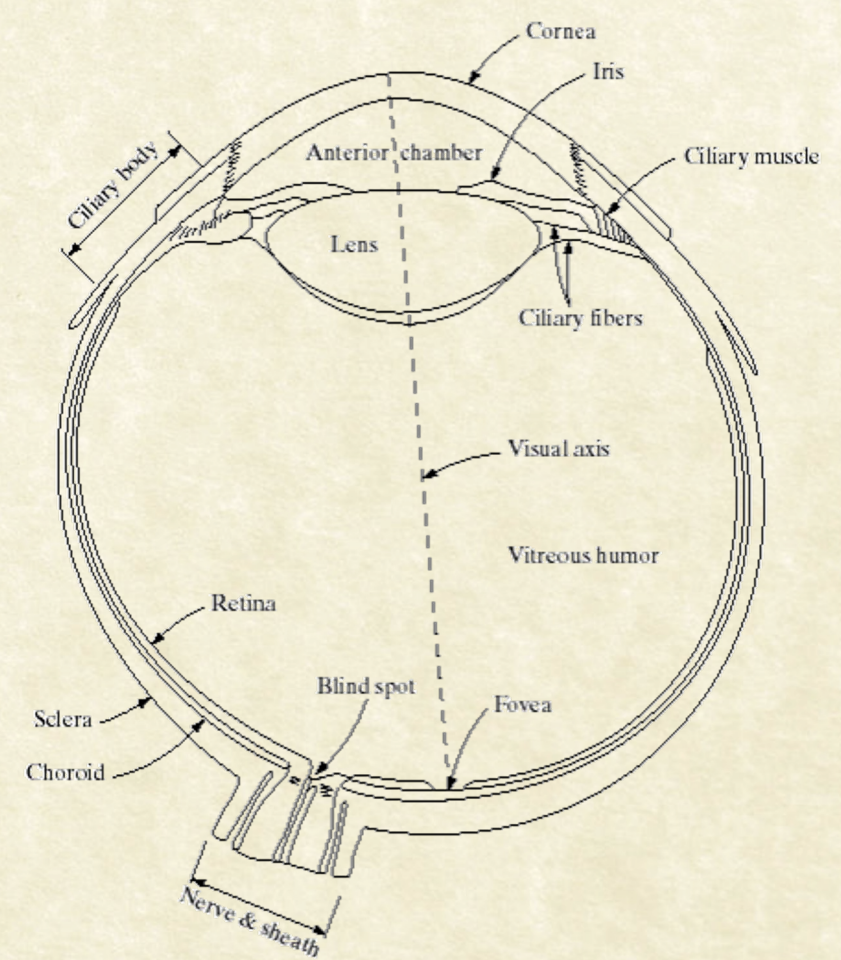
- **Retina:** The imaging surface (not flat)
 - 6 - 7M cones
 - 75 - 150M rods
- **Lens** is a jelly
 - Changes shape to focus
 - Absorbs 8% of VLS; IR, UV
- **Iris** controls the amount of light
 - 2 – 8mm dia.





Cones

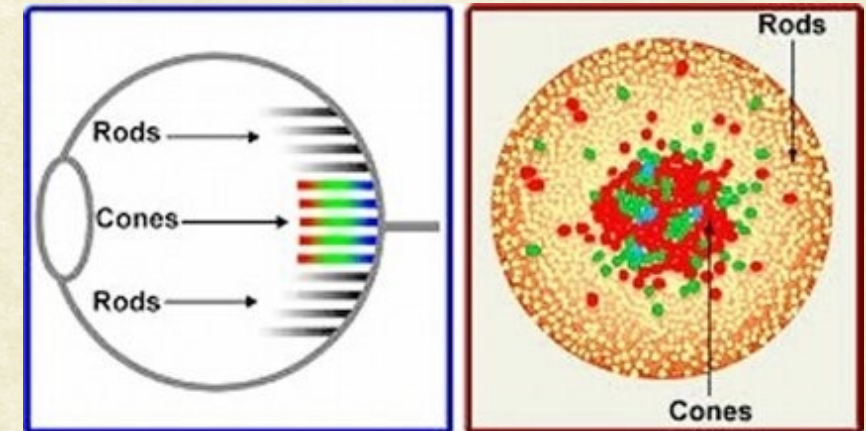
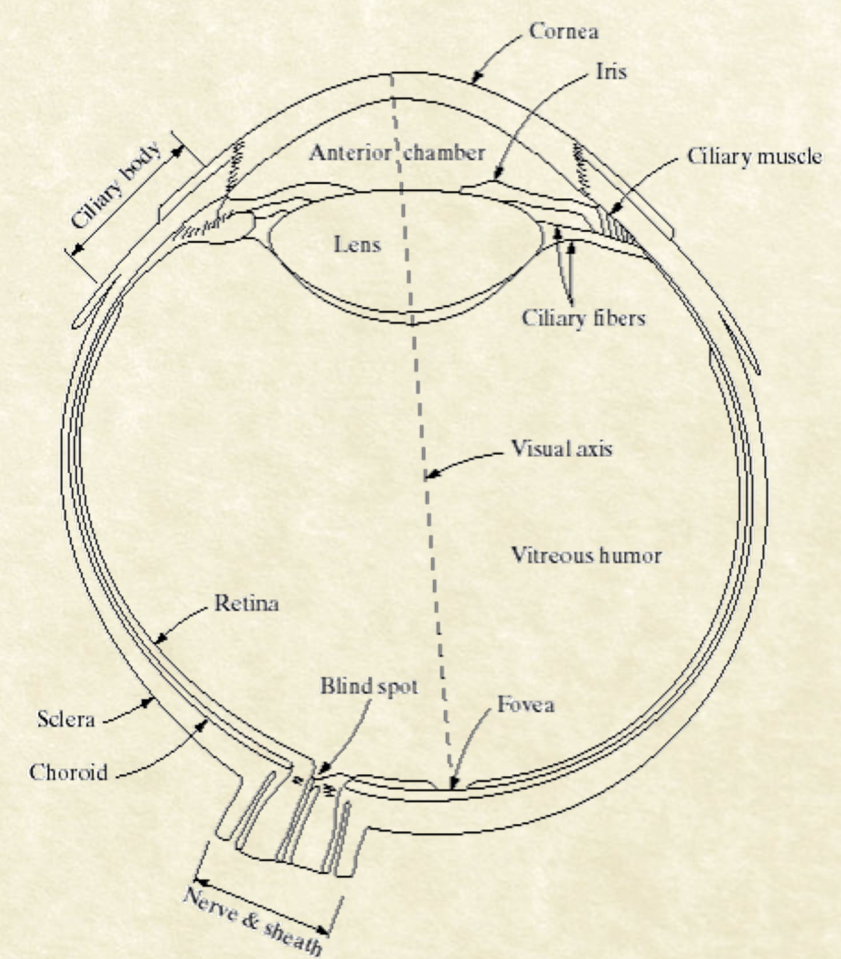
- Located in the fovea
- Sensitive to **color**
- Each cone connected to its own nerve end.
- Sensitive to bright-light: *photopic* vision





Rods

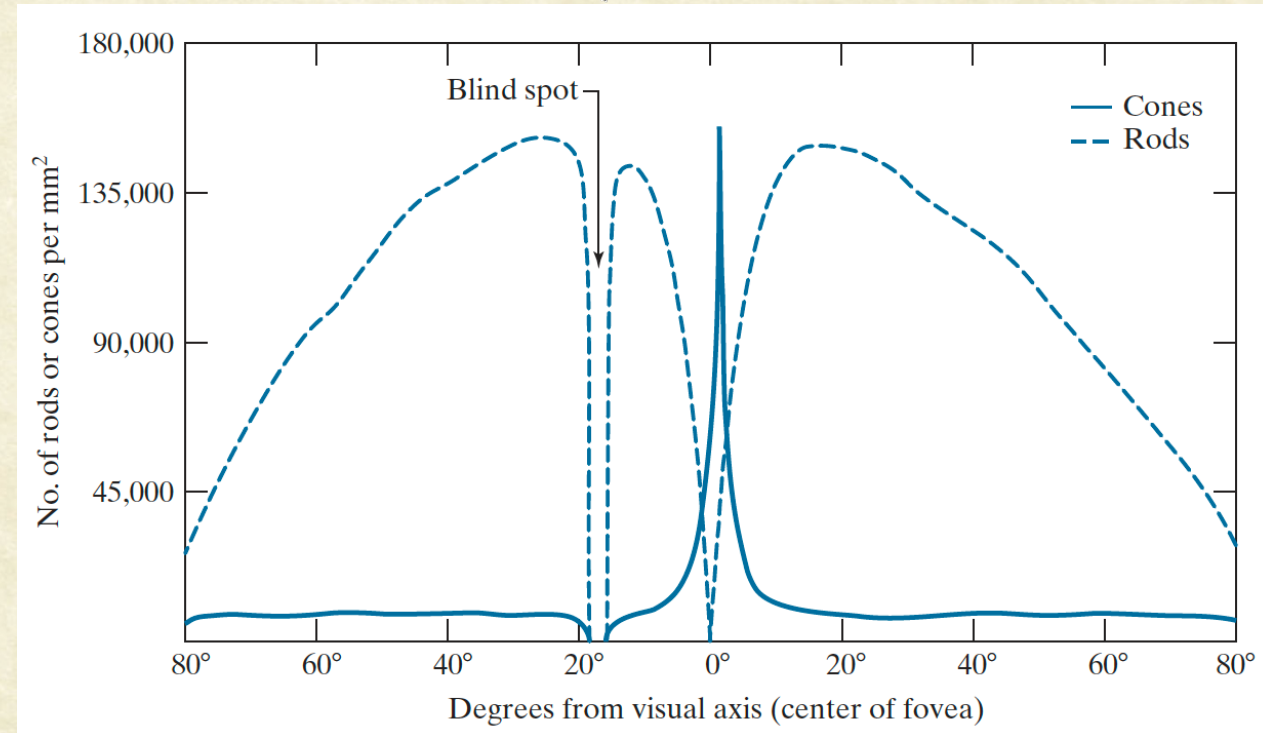
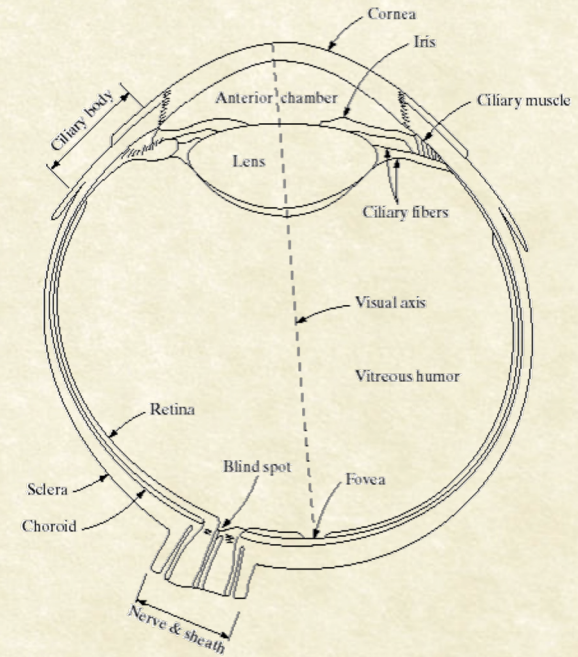
- Provide general, overall picture of the field of view
- Not involved in color vision.
- Several rods are connected to a single nerve.
- Sensitive to low illumination levels (*scotopic* vision).





Receptor Distribution

- Radially symmetric about the fovea.
- Cones are most dense in the center of the fovea
- Rods increase in density from the center to approximately 20° off axis and then decrease.





The Fovea

- Circular (1.5 mm diameter)
 - can be assumed to be a square sensor array (1.5 mm x 1.5 mm).
- Density of cones
 - 150,000 elements/mm² ~ 337,000
 - A CCD imaging chip of medium resolution needs 5 mm x 5 mm for this number of elements

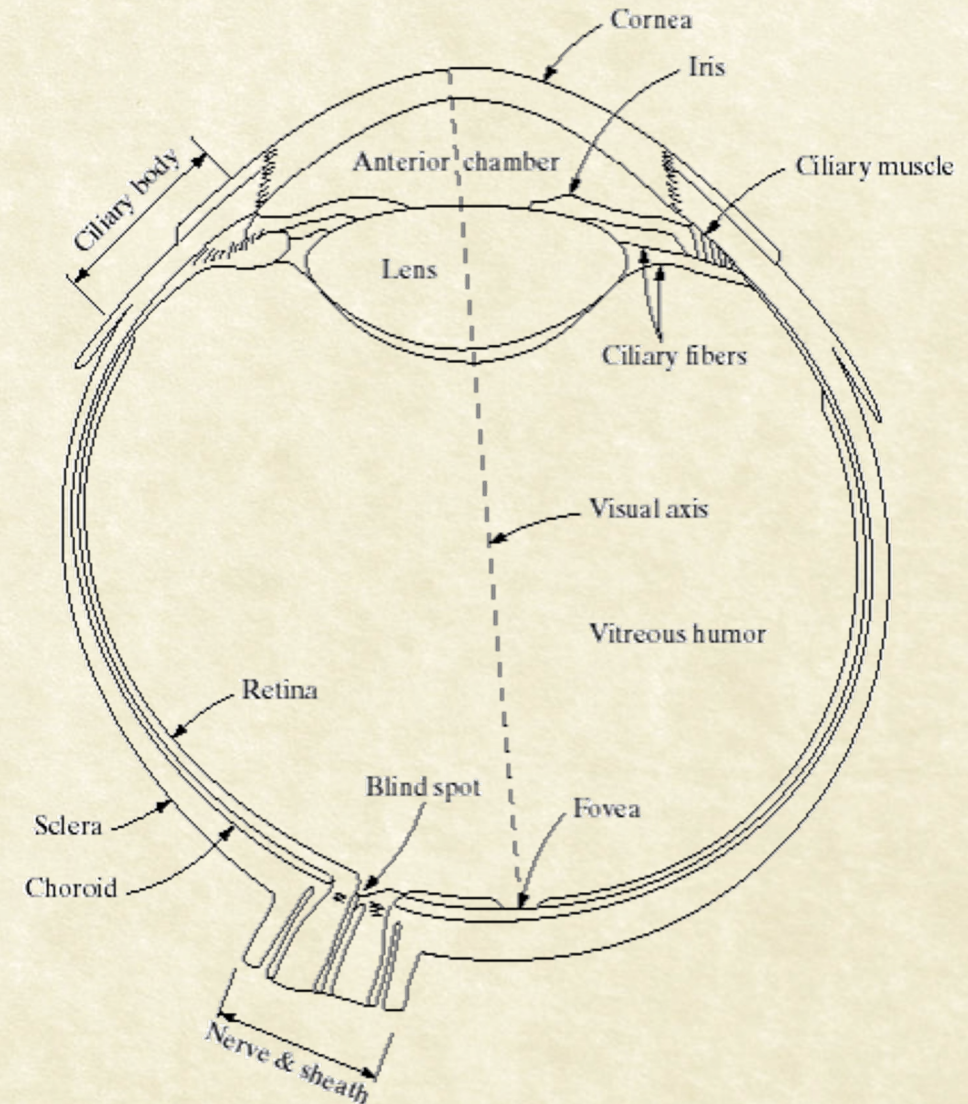
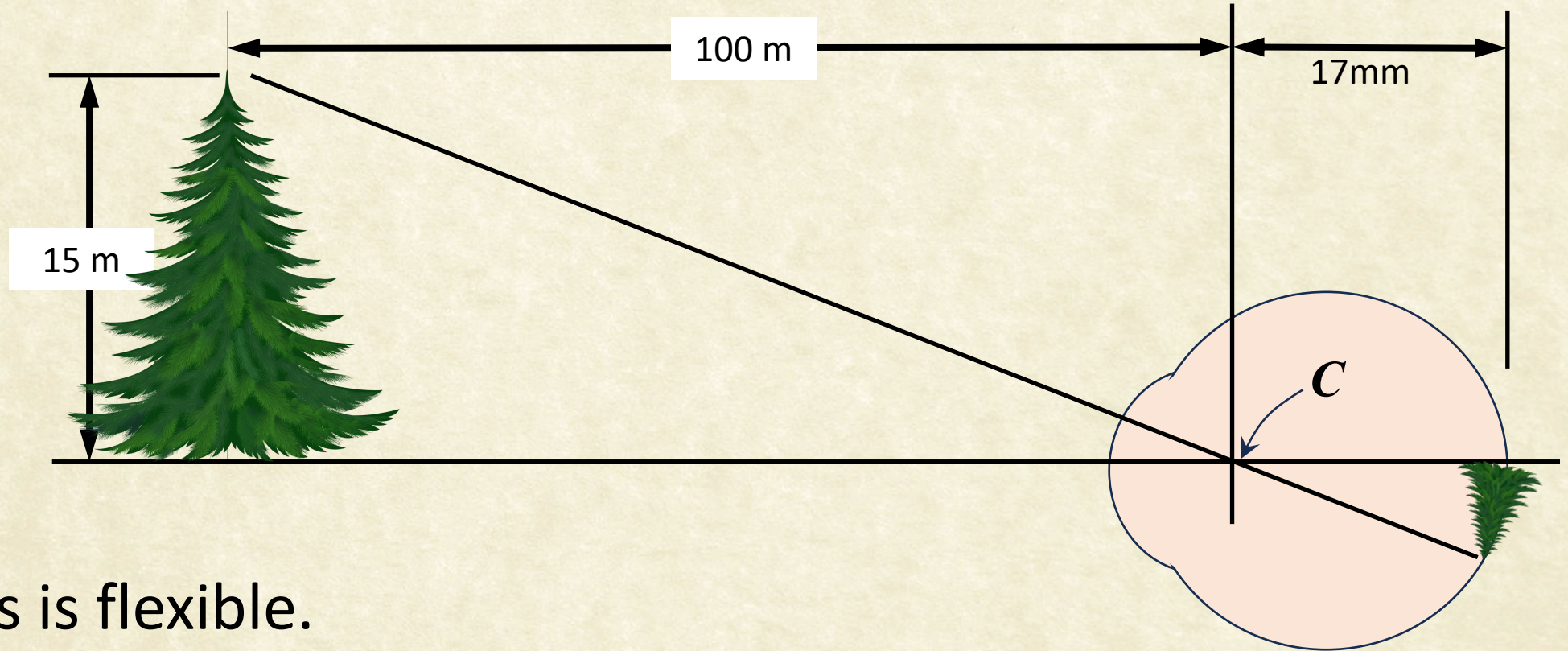




Image Formation in the Eye

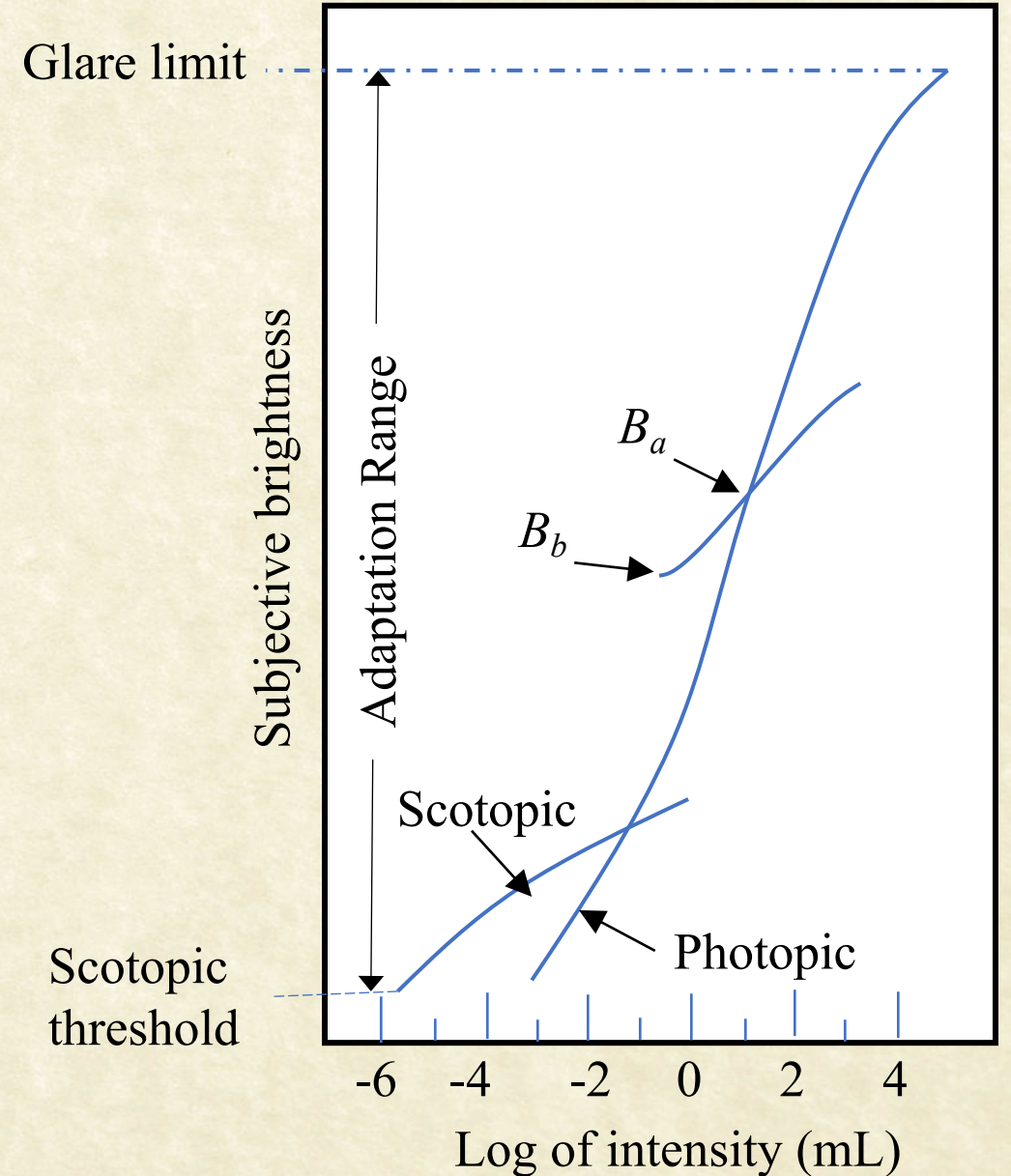


- Eye lens is flexible.
- Lens controlled by fibers of ciliary body
 - To focus on distant objects, it gets flatter (and vice versa)
 - Focal length varies from 14-17 mm



Brightness Adaptation

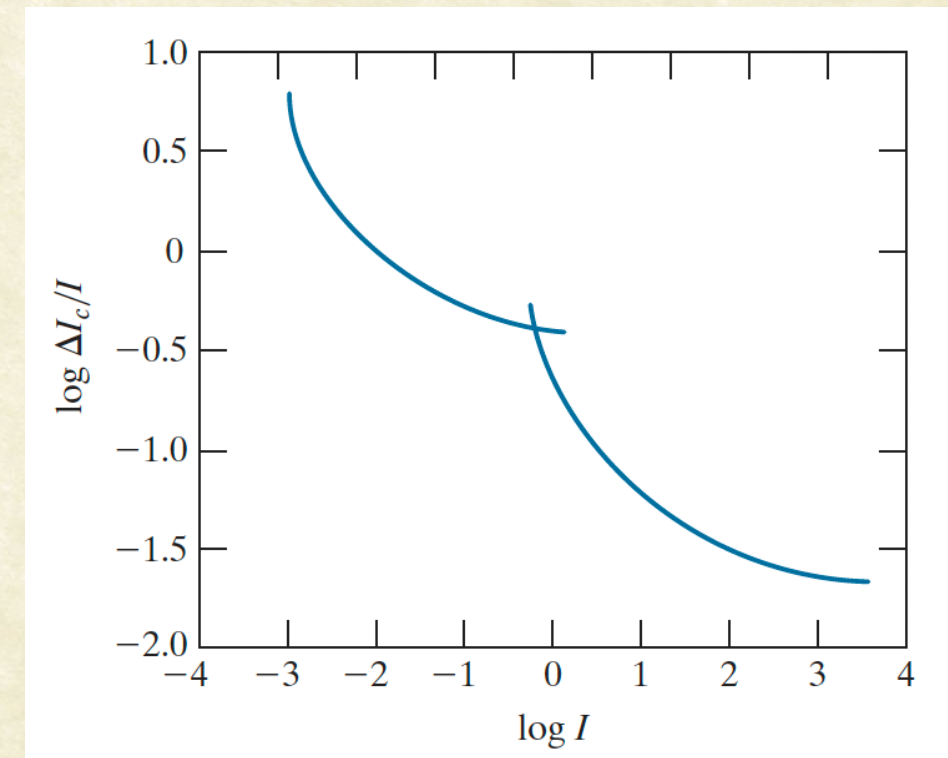
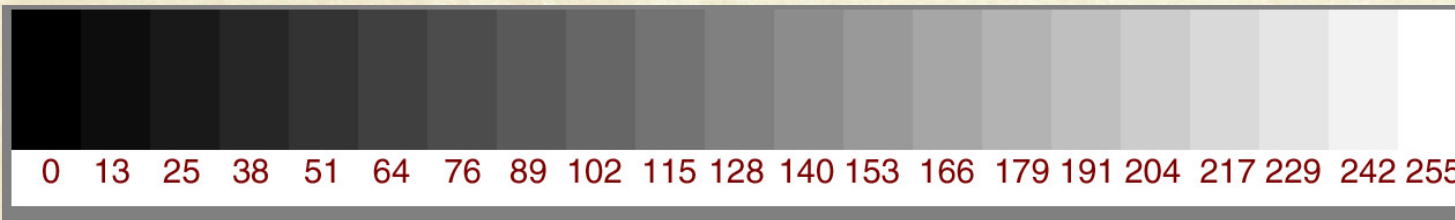
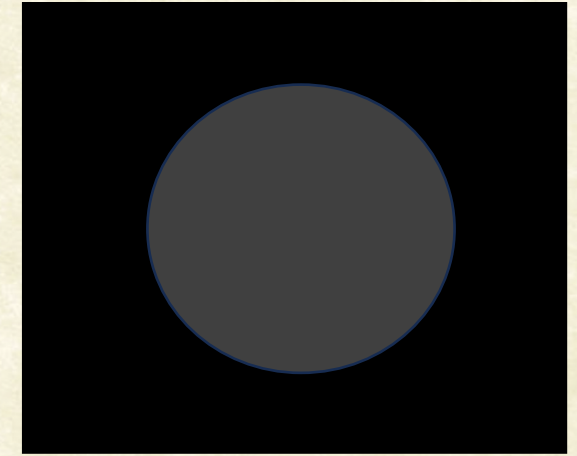
- Dynamic range of human visual system (HVS)
 - $10^{-6} \sim 10^4$
- But HVS cannot accomplish this range simultaneously





Brightness Discrimination

- Weber ratio: $\Delta I_c / I$
 - I : the background illumination
 - ΔI_c : the increment of illumination
- Small Weber ratio $\rightarrow$ good discrimination
- Larger Weber ratio $\rightarrow$ poor discrimination

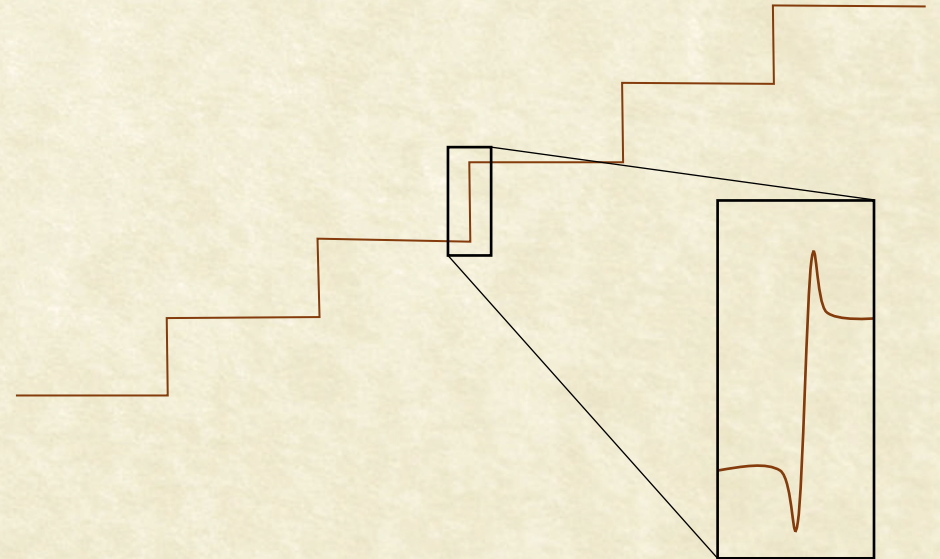
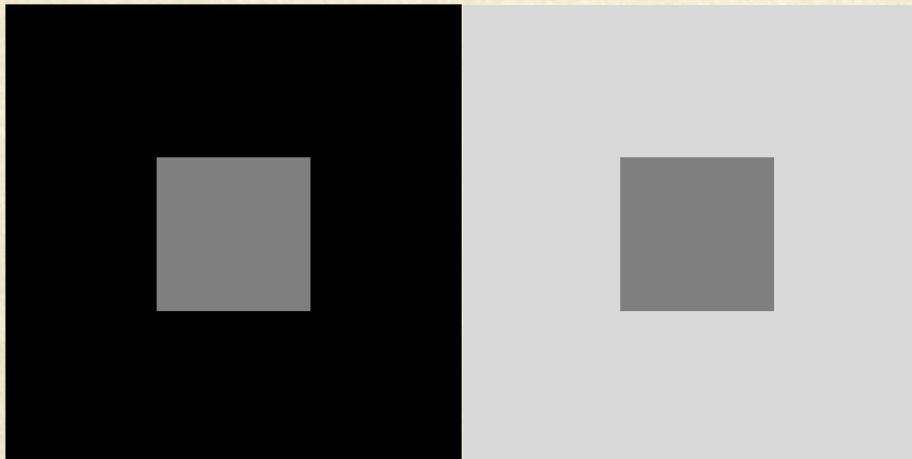




Psychovisual Effects

Perceived brightness and contrast are not a simple function of intensities:

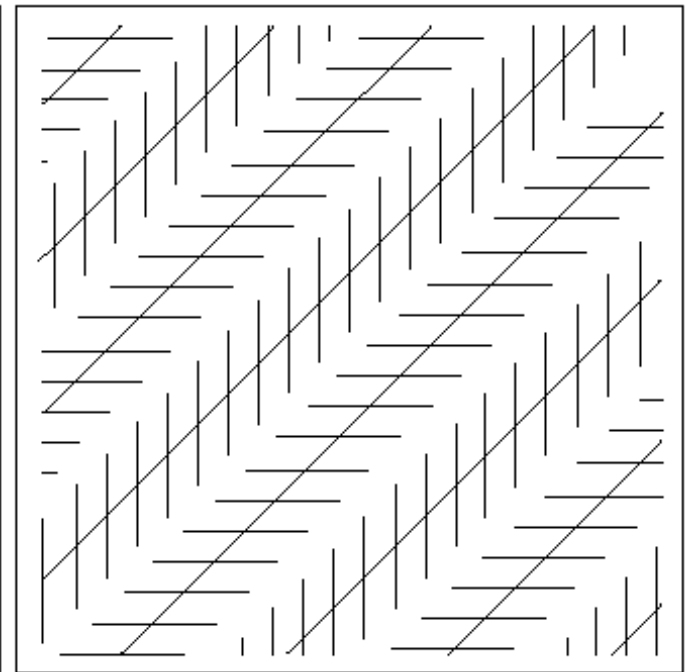
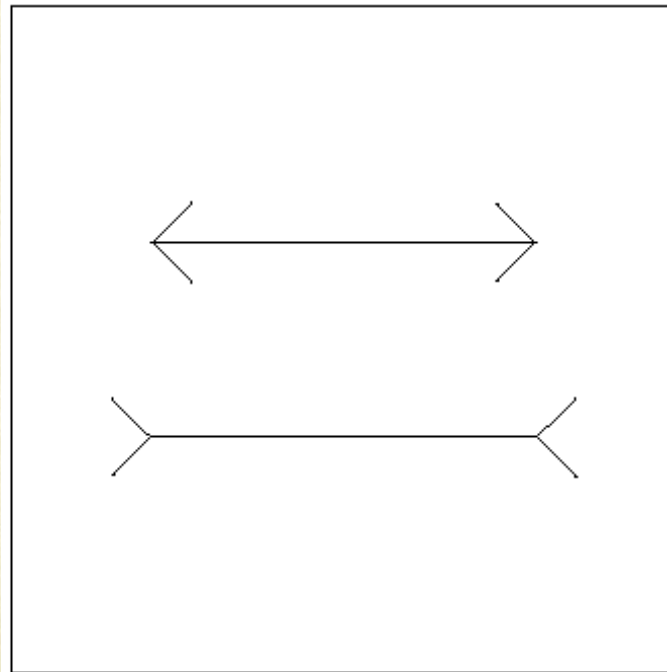
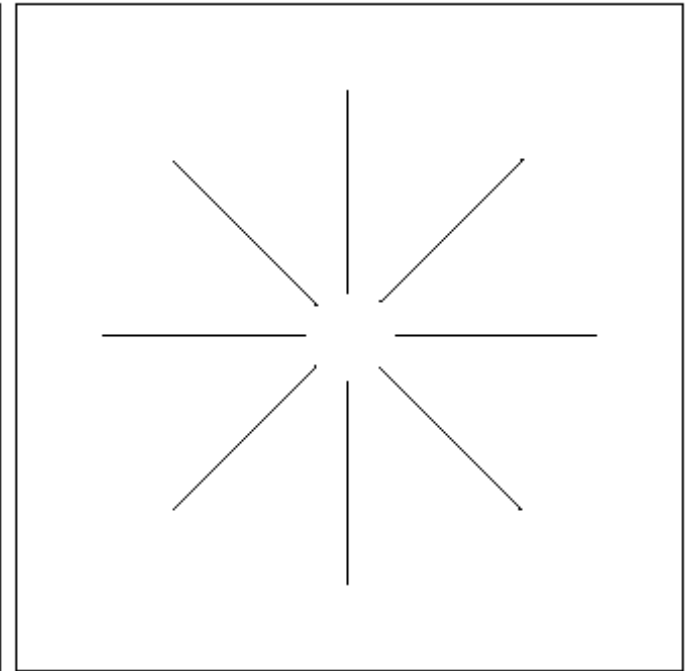
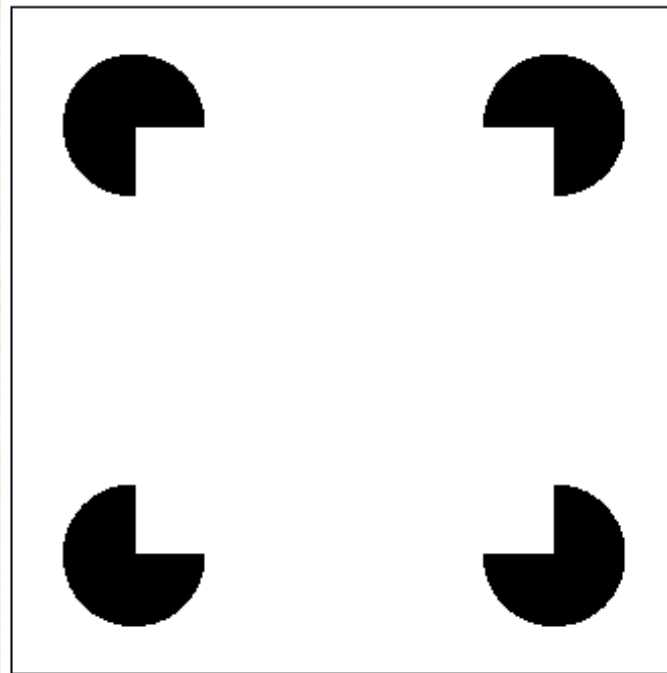
1. Simultaneous Contrast
2. Mach Band Effect





Optical Illusions

We perceive edges,
incidence, length,
parallelism, all in
context.





Questions

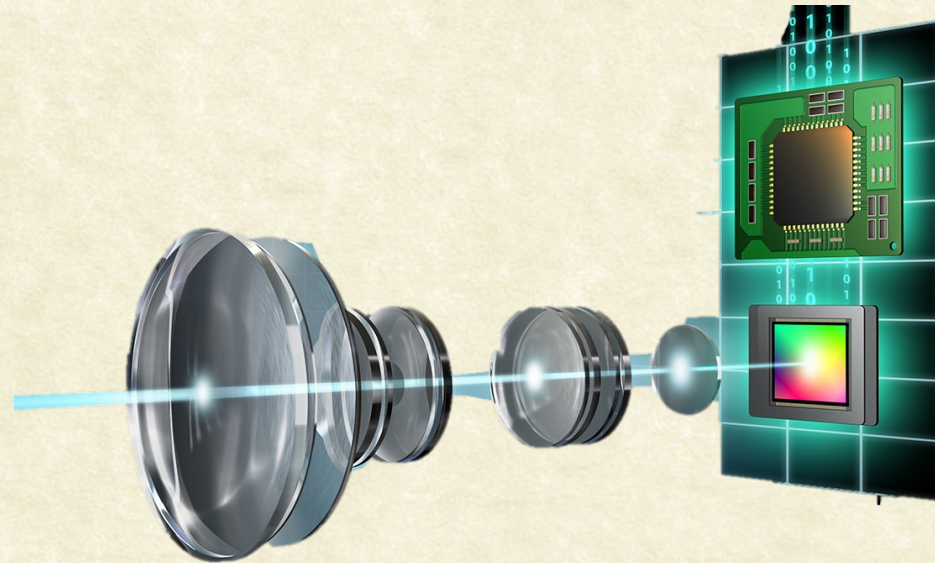


Questions?



CS7.404: Digital Image Processing

Monsoon 2025: II: Image Acquisition

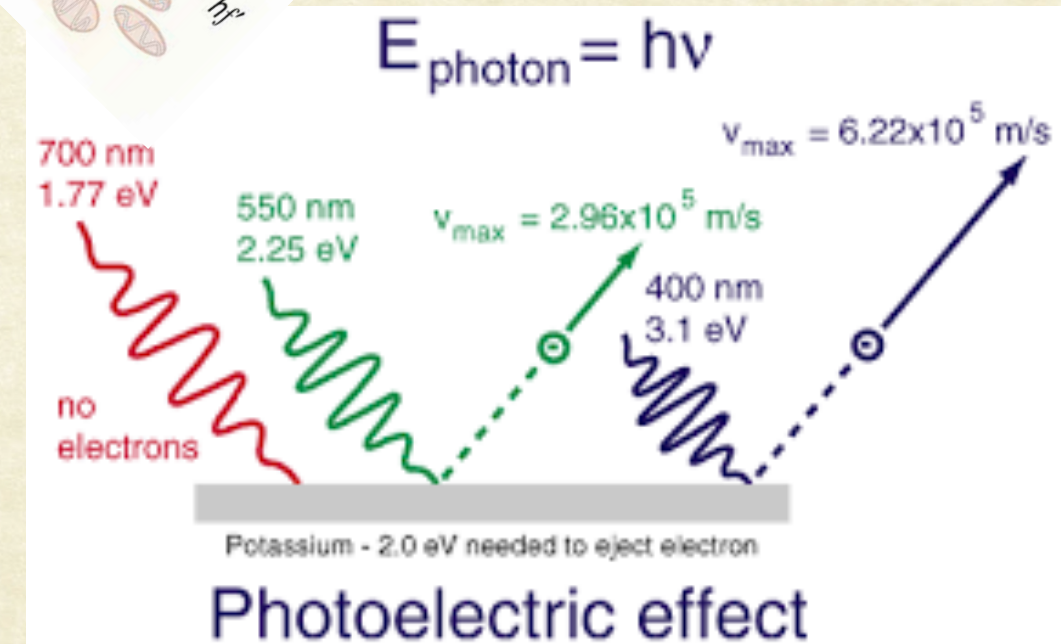
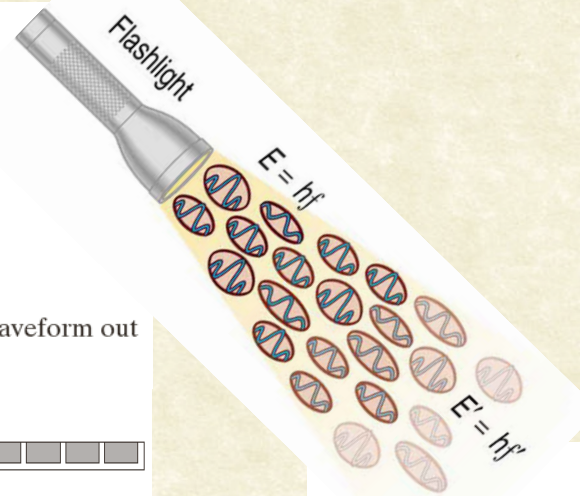
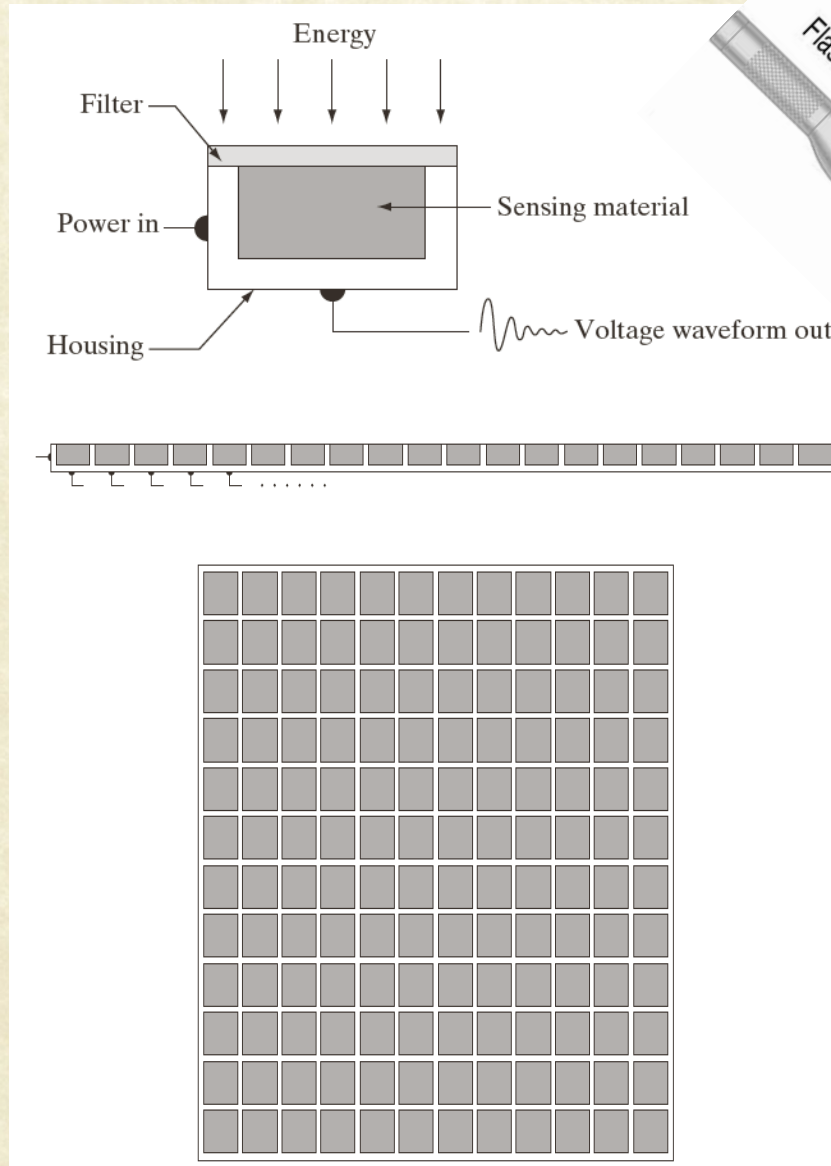


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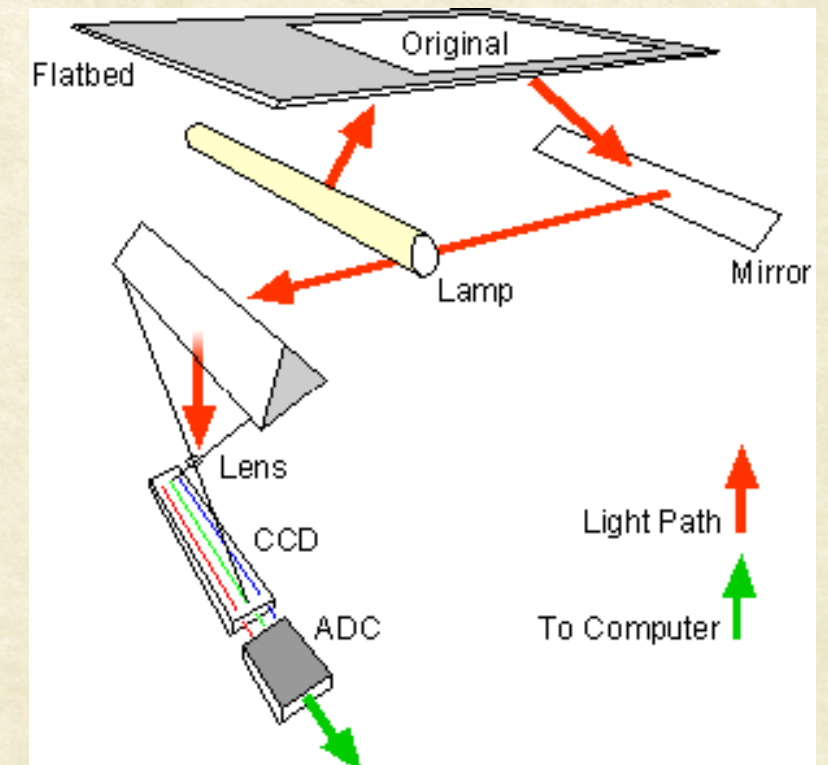
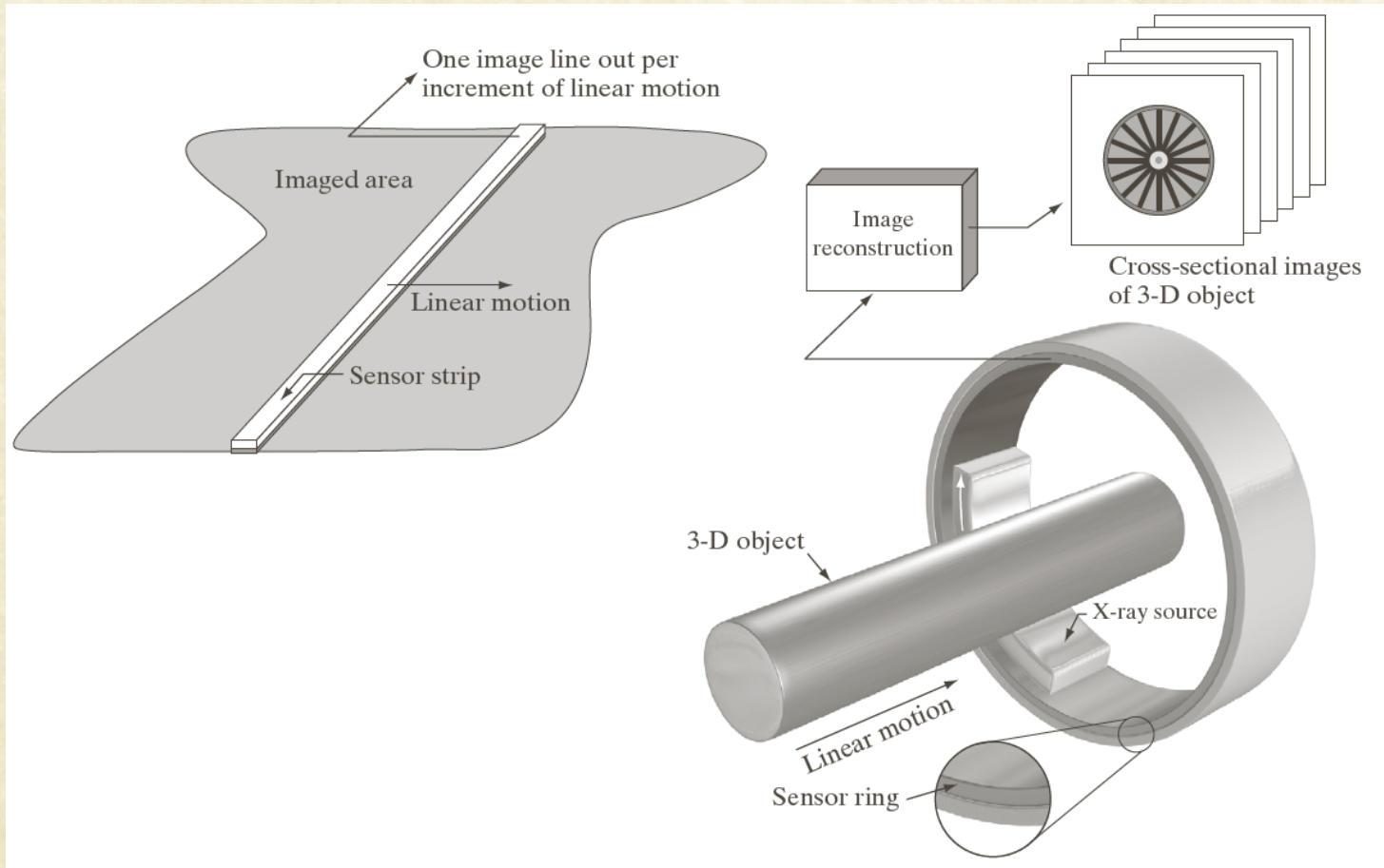
Image Sensing and Acquisition

- Single Sensor
- Line Sensor
- Array Sensor



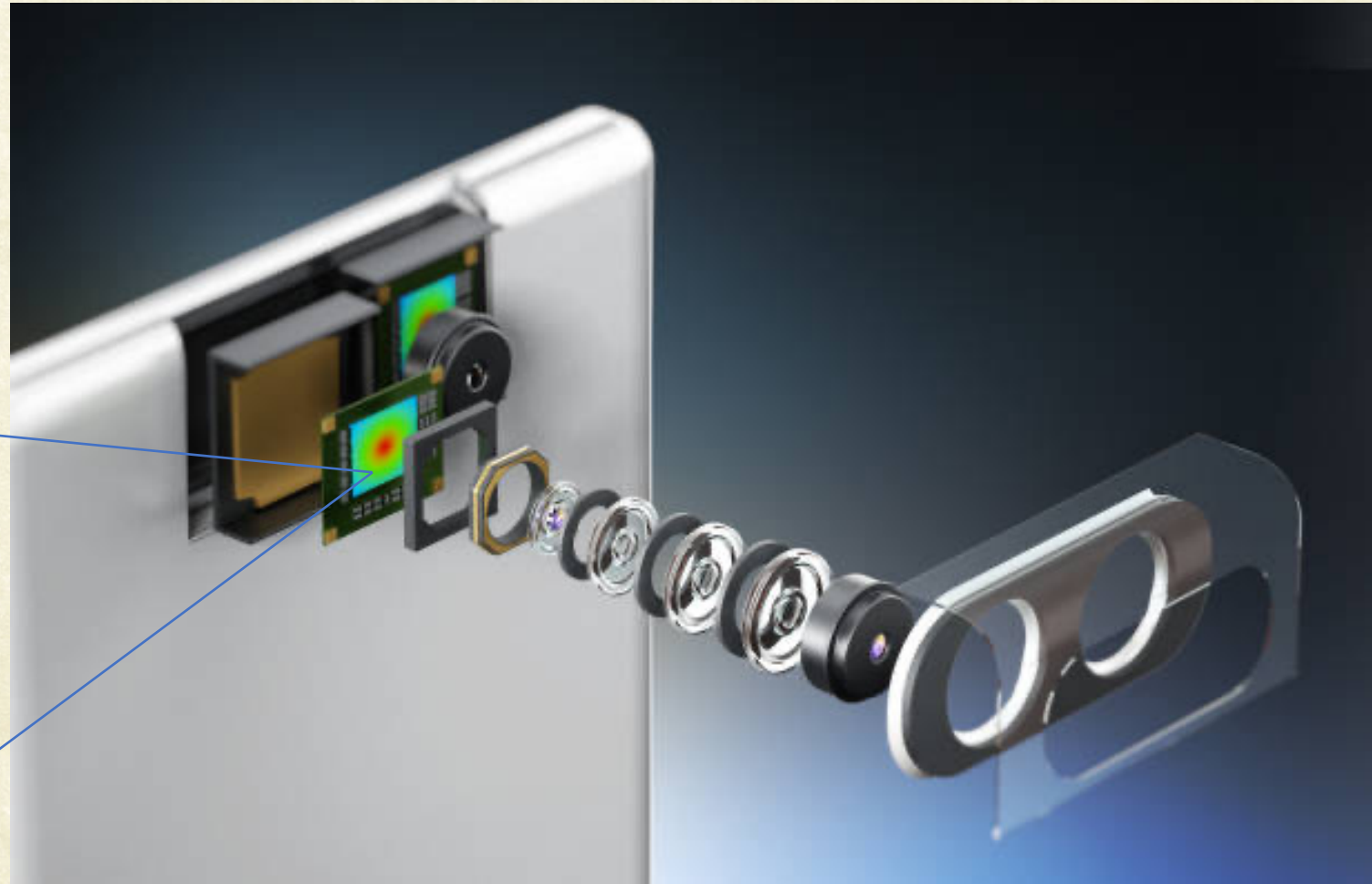
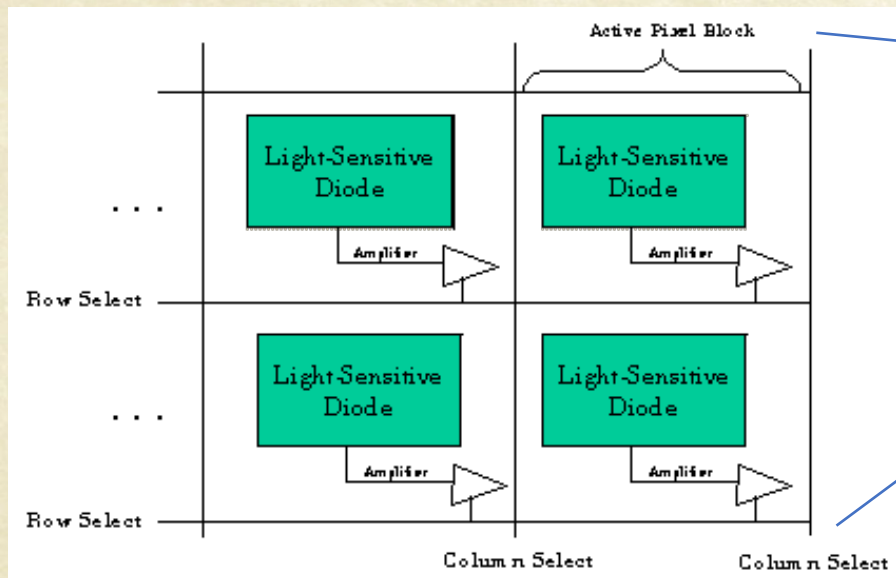
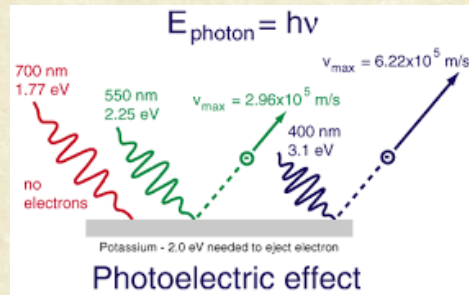


Line Sensors





CMOS photo-electric sensor





Light → Color

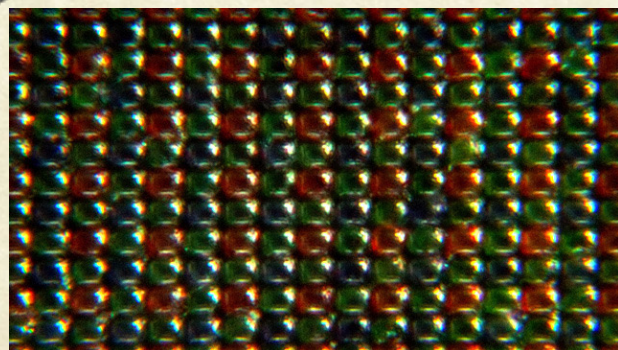
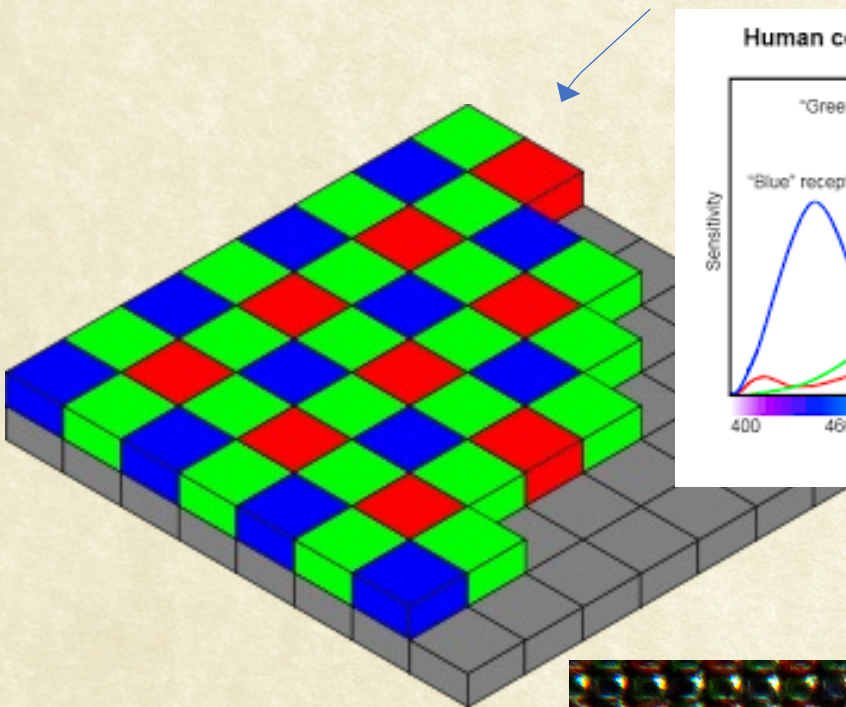
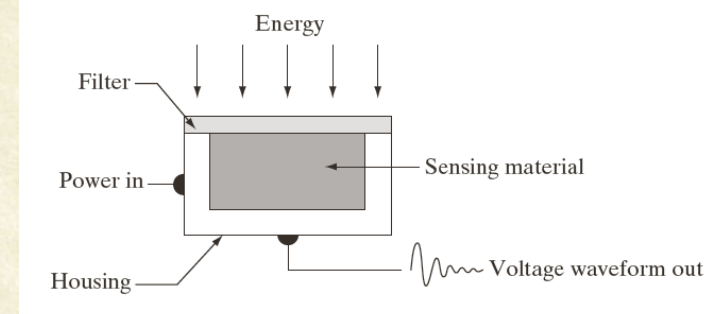
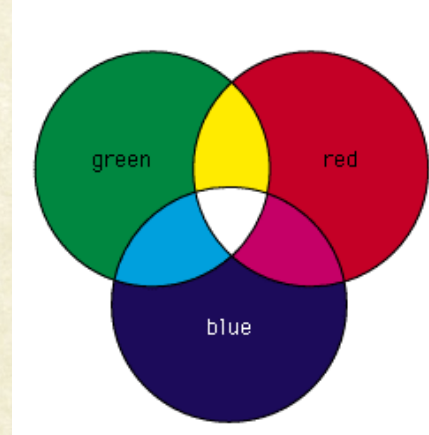
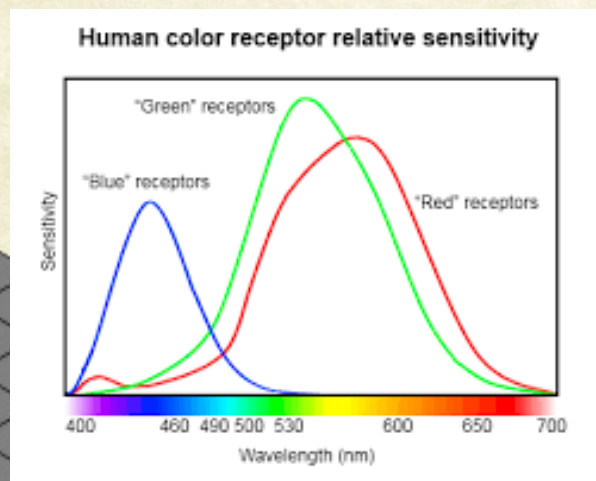
- CMOS sensitive to “light”, not “color”



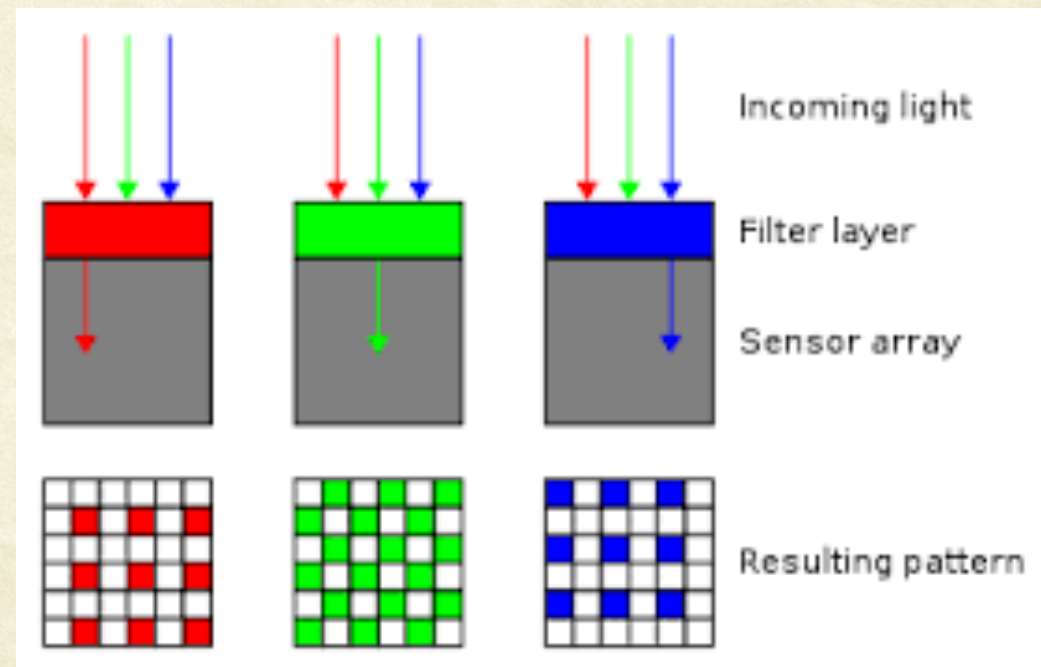


Bayer Filter

Relatively more green filters. Why ?

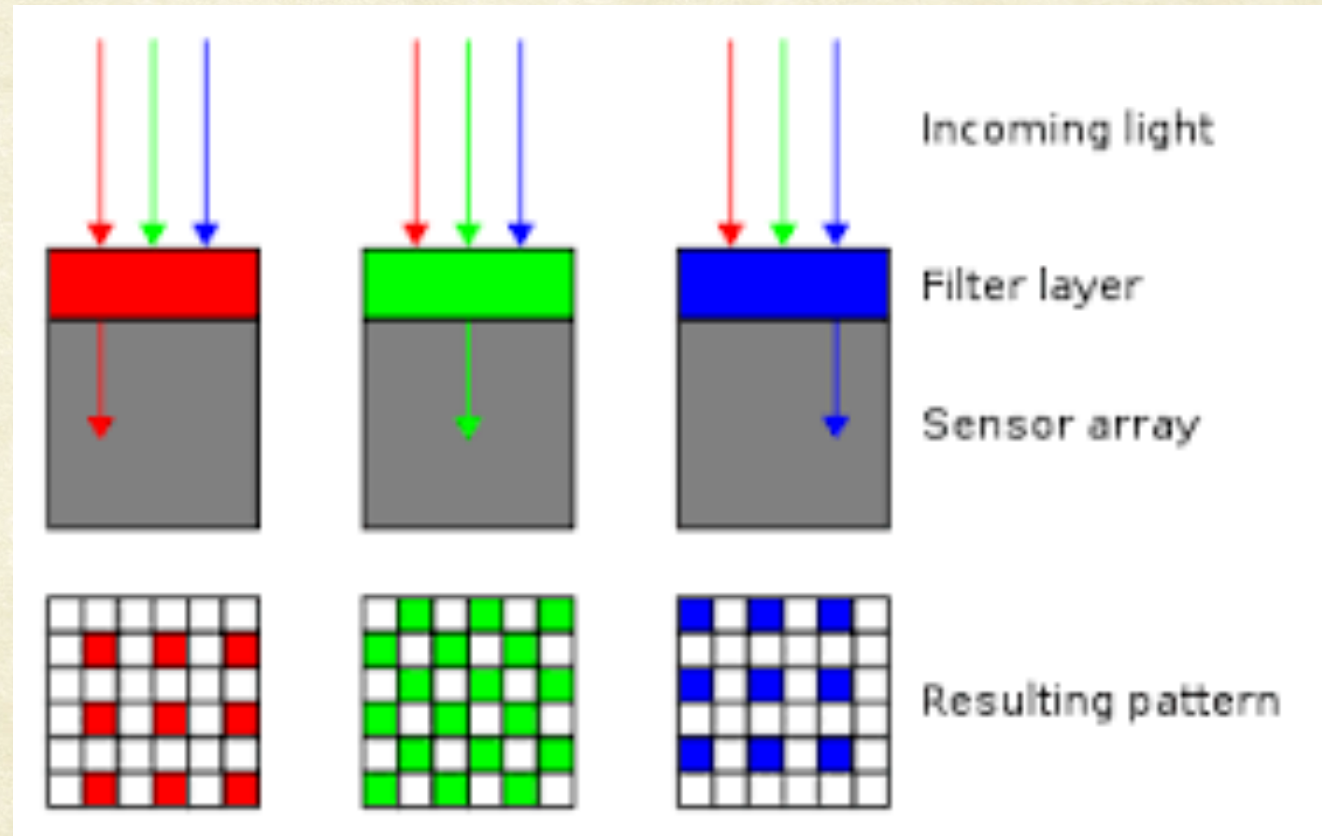
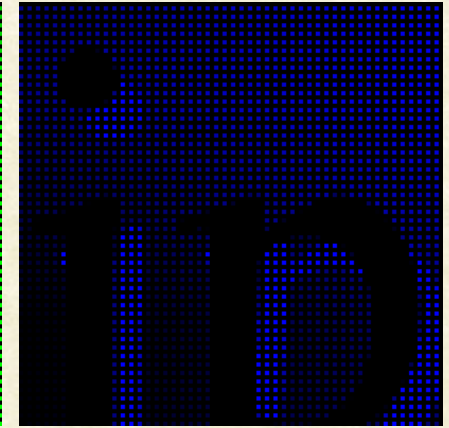
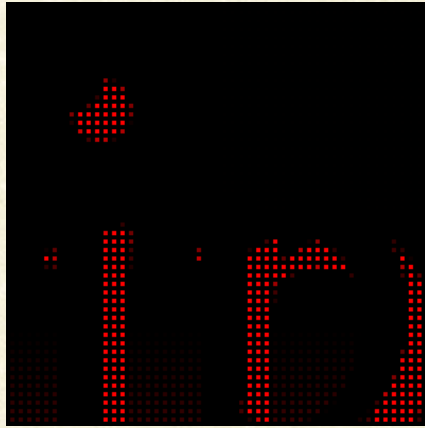


600x



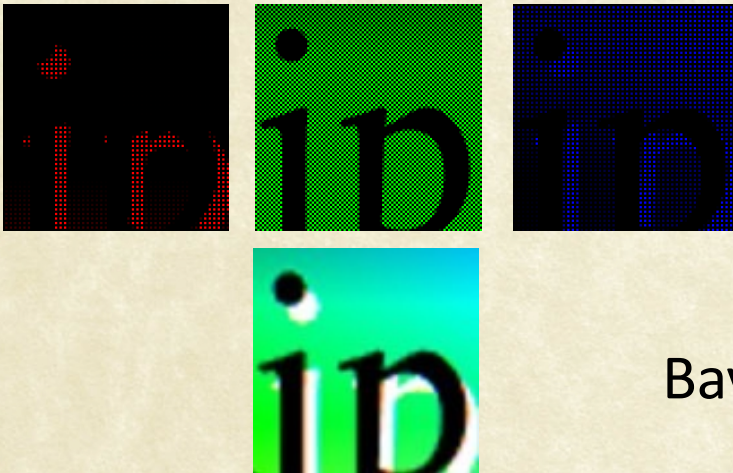
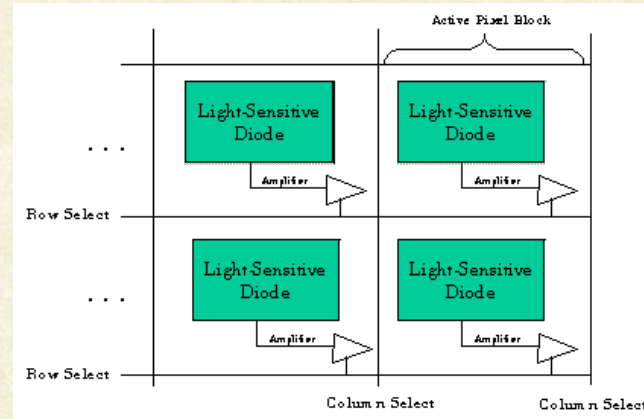
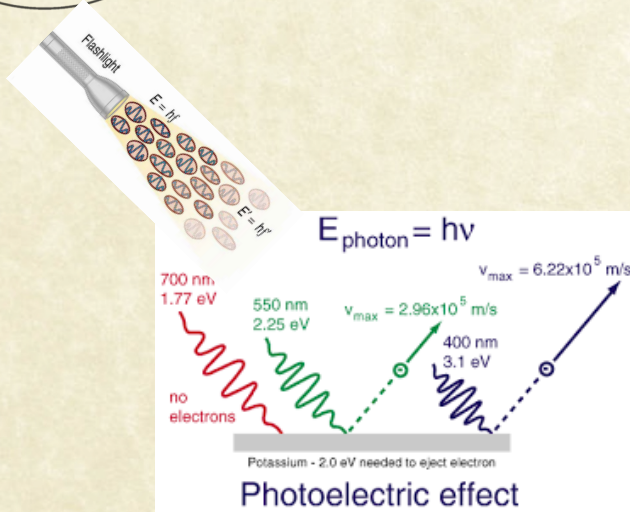


Demosaicing

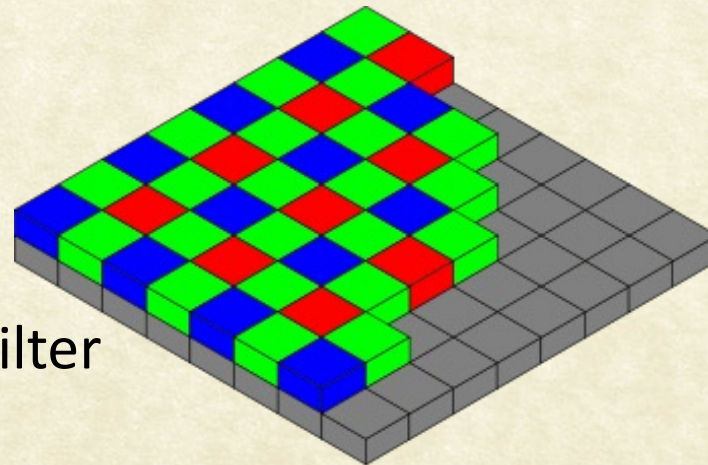




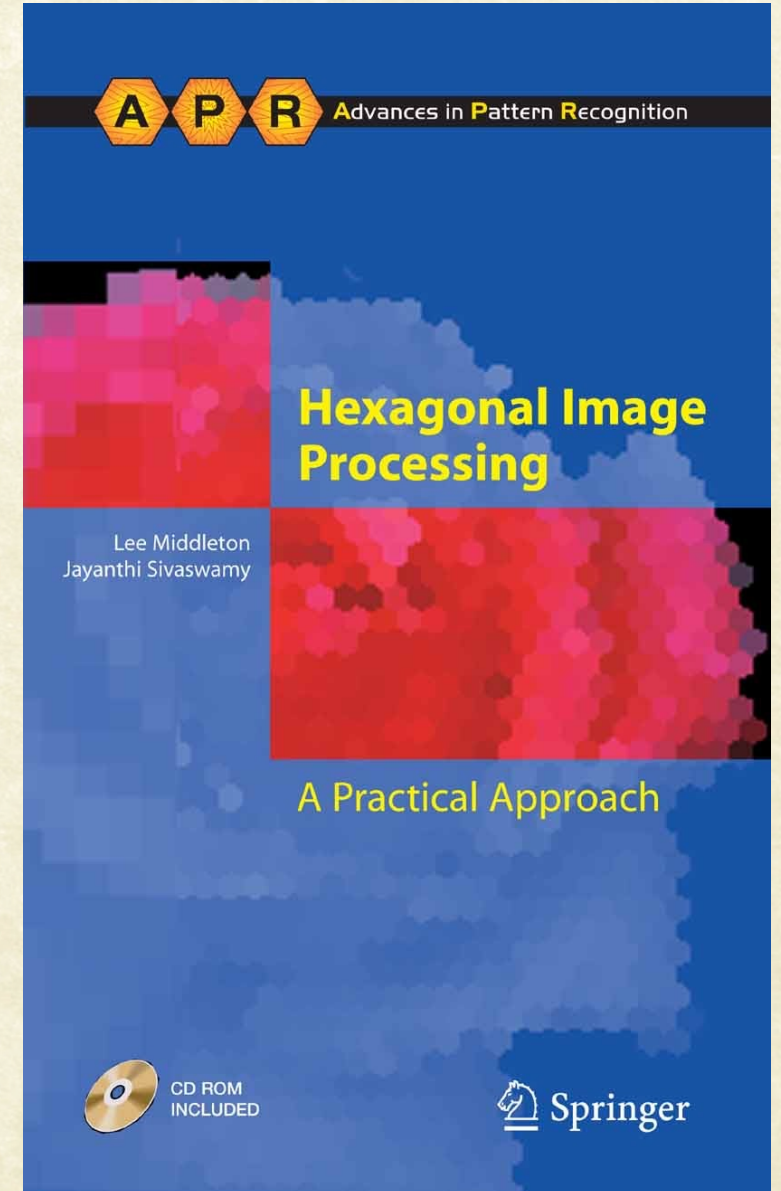
Imaging Summary



Demosaicing



Bayer Filter



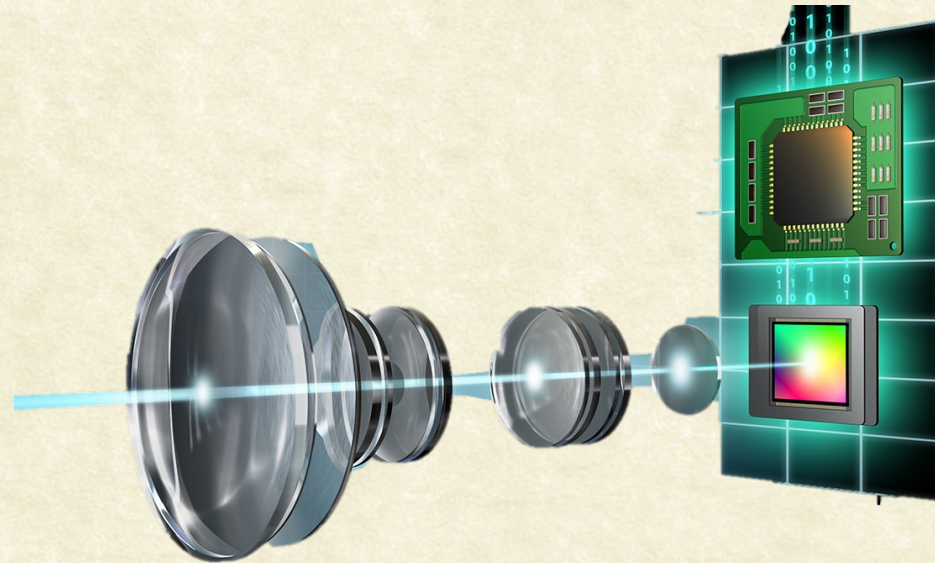


Questions?



CS7.404: Digital Image Processing

Monsoon 2025: III: Image Signals

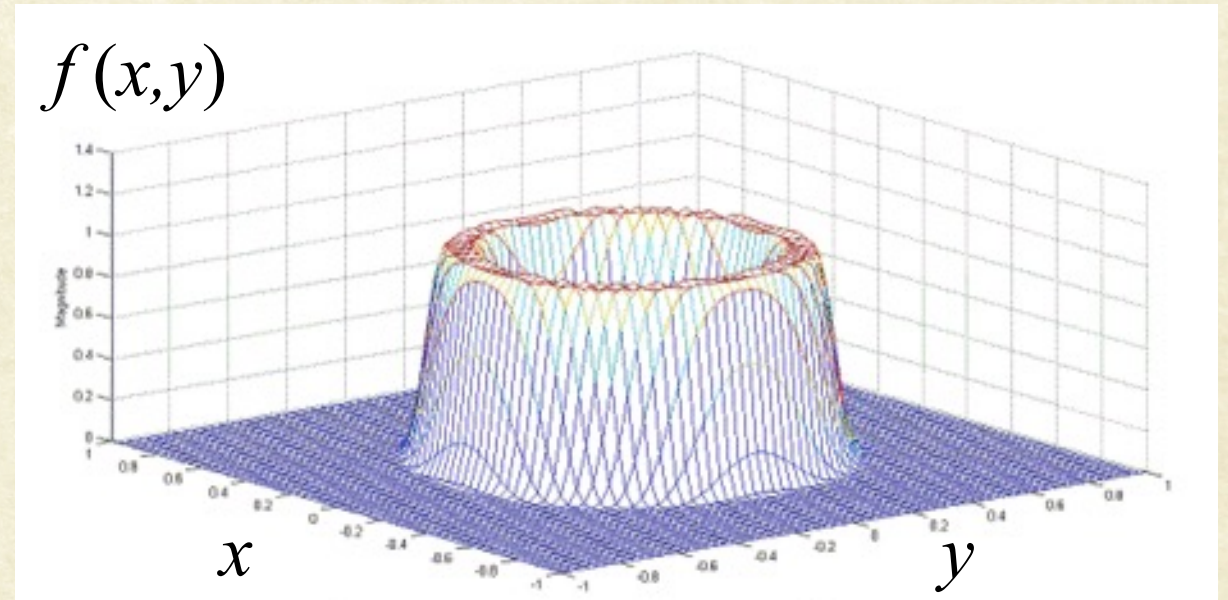
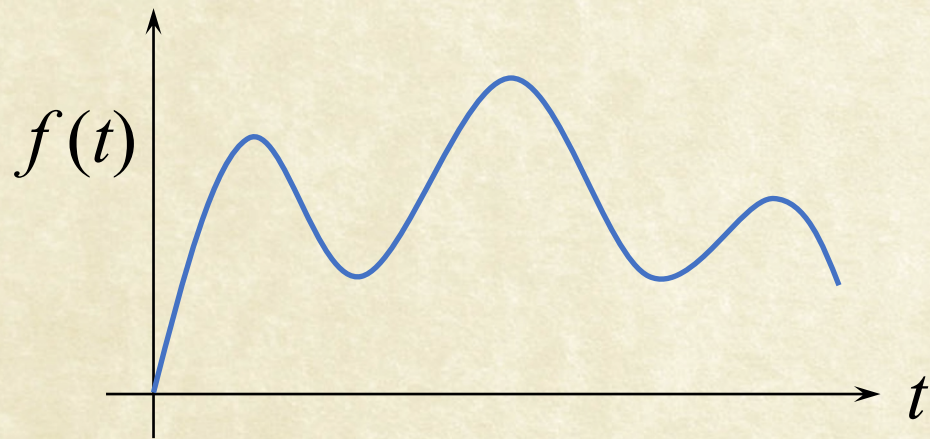
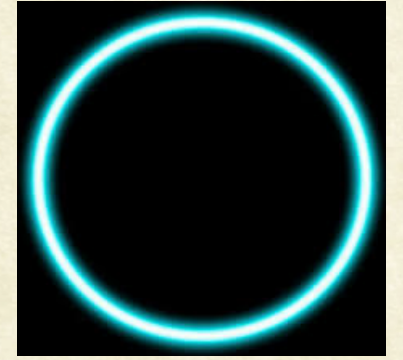


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Signal

”A function that conveys information about the behavior or attributes of some phenomenon”



Analog Signal



Analog vs. Digital Signal (2D)

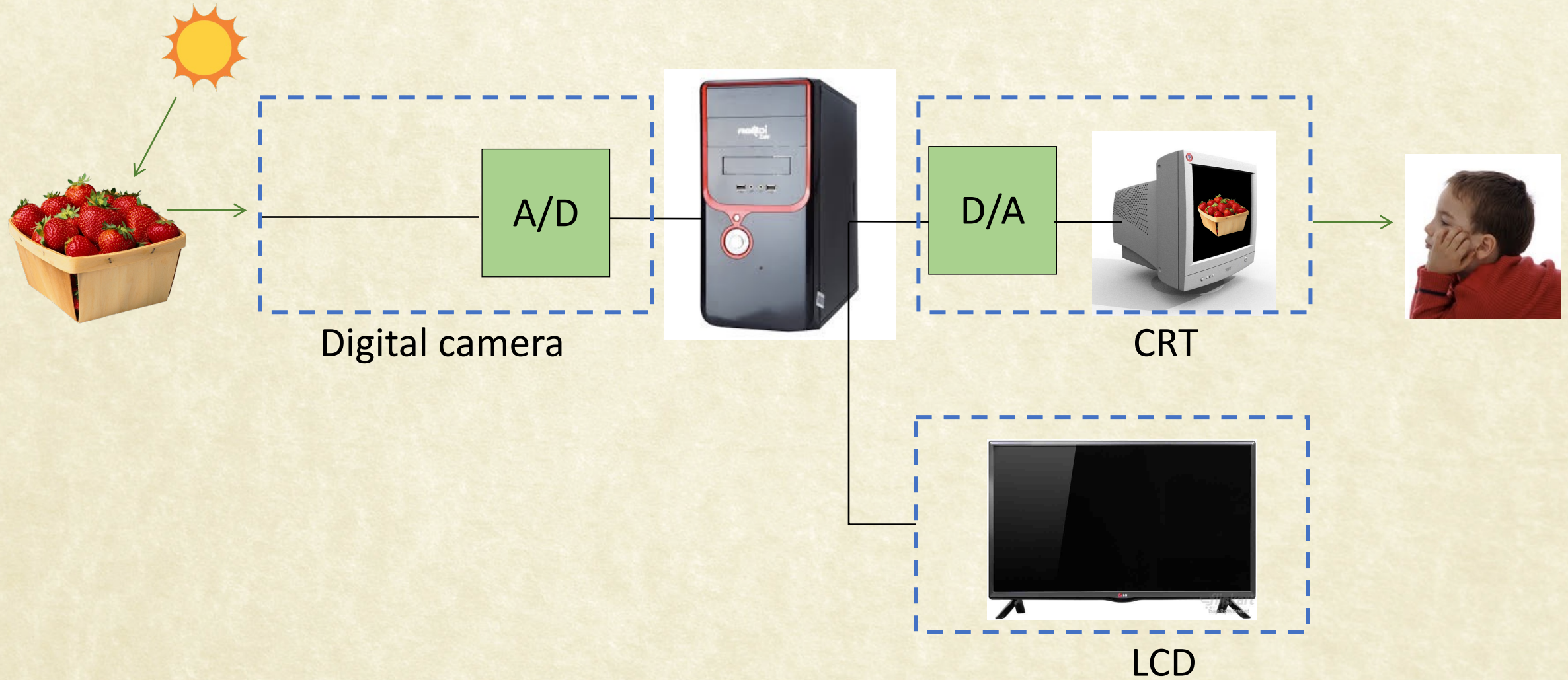
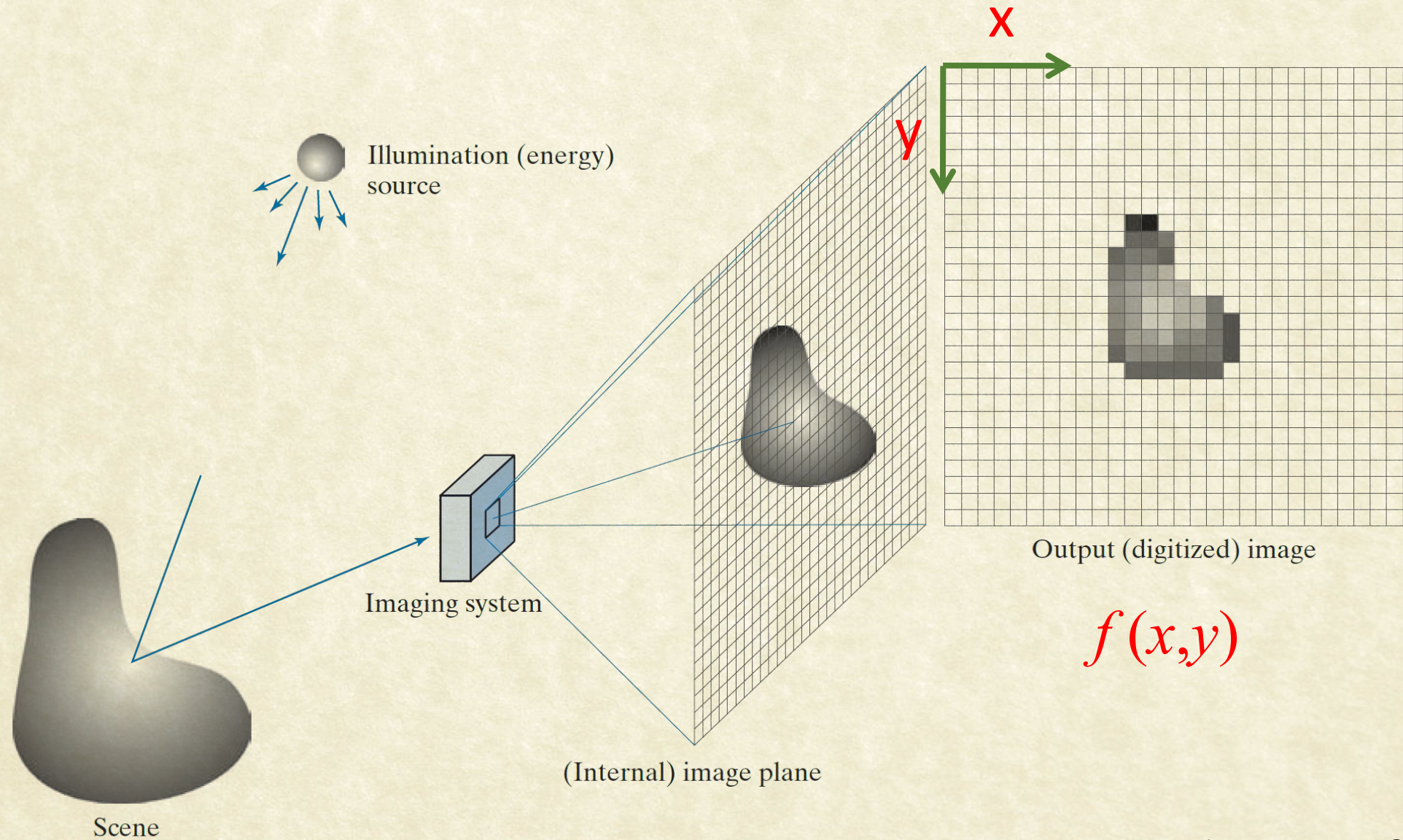
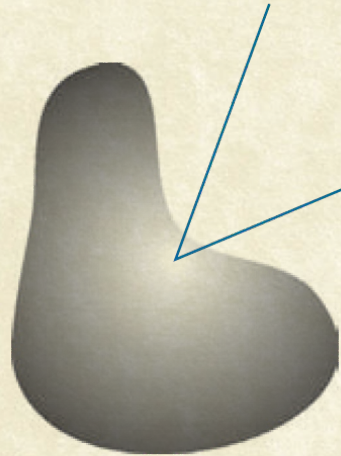
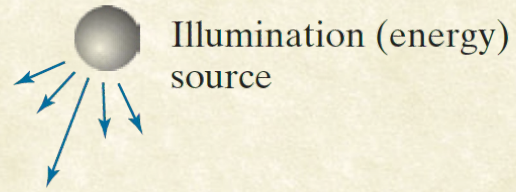


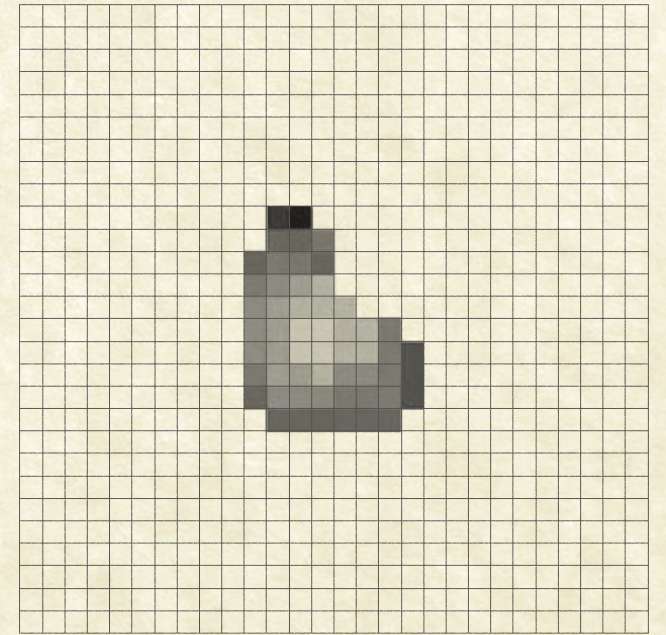
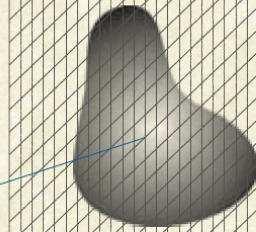


Image Acquisition Process

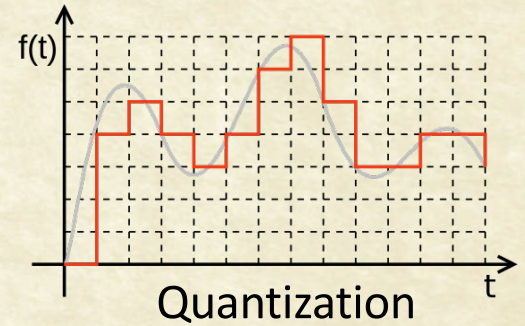
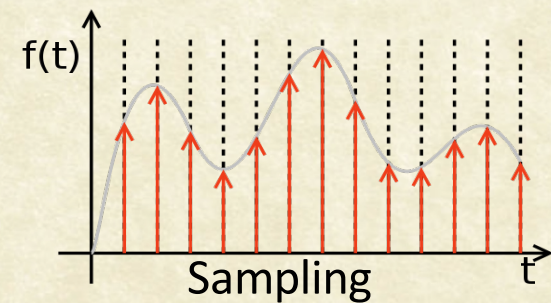
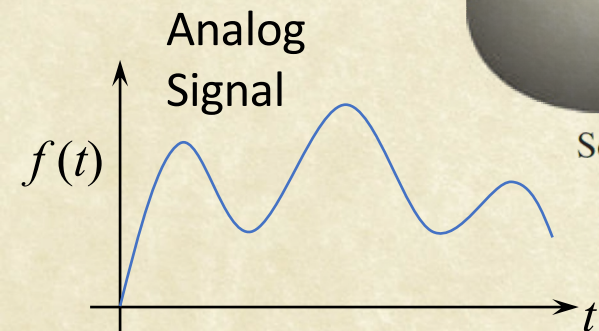




Imaging system



Output (digitized) image





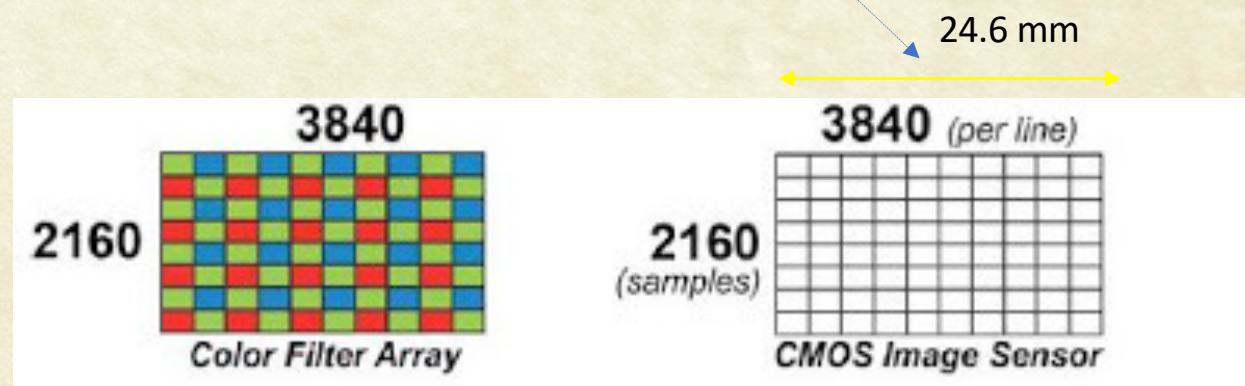
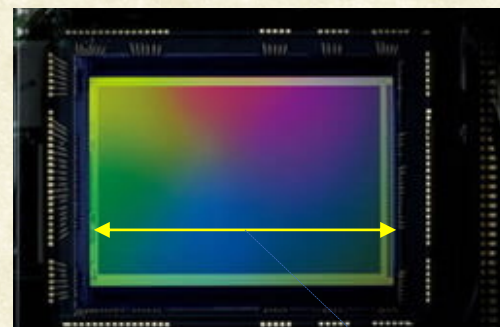
Cross-section of typical smartphone camera



Physical Characteristics

Active image area size	24.6 (H) x 13.8 (V) mm
Total number photosites	4206 (H) x 2340 (V)
Number photosites for active image	3840 (H) x 2160 (V)
Color filter array (with microlens)	RGB Bayer
Size of photosite (microns)	6.4 (H) x 6.4 μm
Pixel pitch	6.4 μm
Power supply	3.3v / 1.8v
Power consumption	950mW

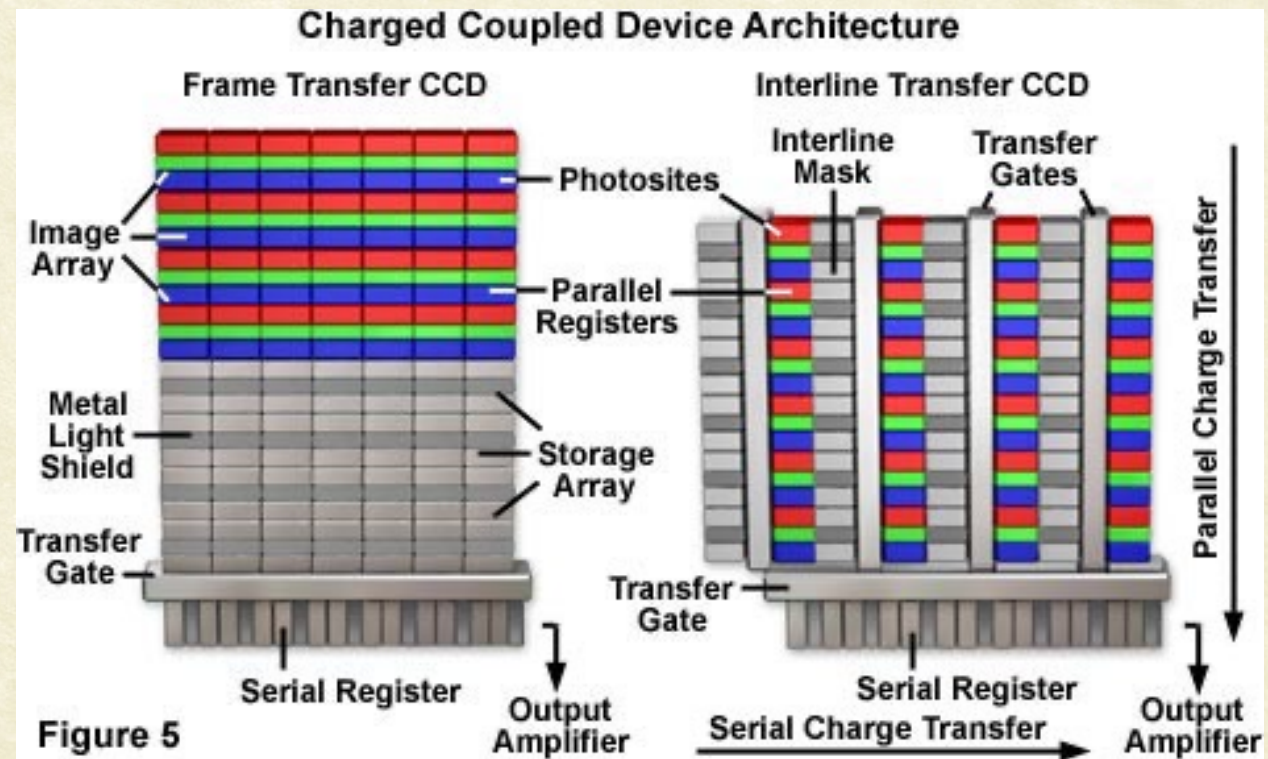
Resolution (of the sensor)





CCD and CMOS cameras

- Non-Bayer Arrangements





Sampling

- Grayscale Image

A two-dimensional function $f(x,y)$ of real values on the continuous image plane.

- Sampling

The process of converting a continuous image $f(x,y)$ to a discrete representation $f_s[m,n]$ by recording the function values at points of intersection of a 2D grid.

The process is equivalent to multiplying the image function with an ideal 2D impulse train $\delta(x,y)$. X_0 and Y_0 are called sampling distances or intervals.

$$\begin{aligned} f_s[m,n] &= f(x,y) \cdot \sum_{m=-\infty}^{+\infty} \sum_{n=-\infty}^{+\infty} \delta(x - mX_0, y - nY_0) \\ &= \sum_{m=-\infty}^{+\infty} \sum_{n=-\infty}^{+\infty} f(mX_0, nY_0) \delta(x - mX_0, y - nY_0) \end{aligned}$$



Sampling

Image function

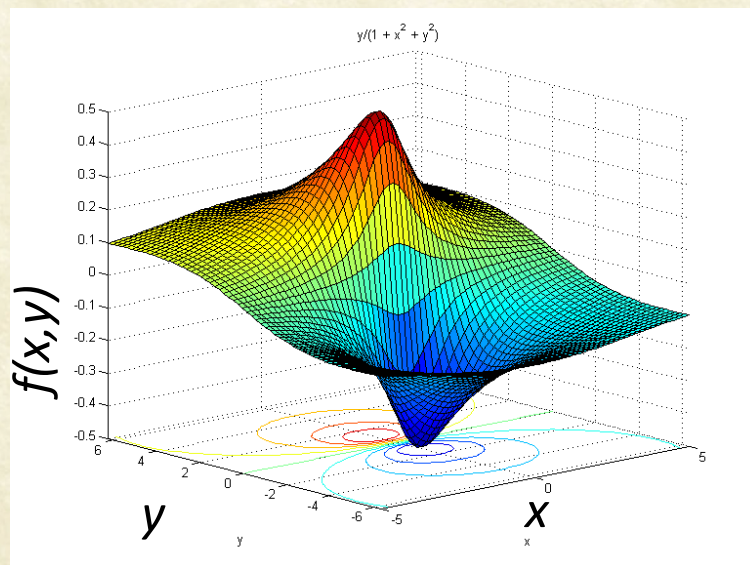
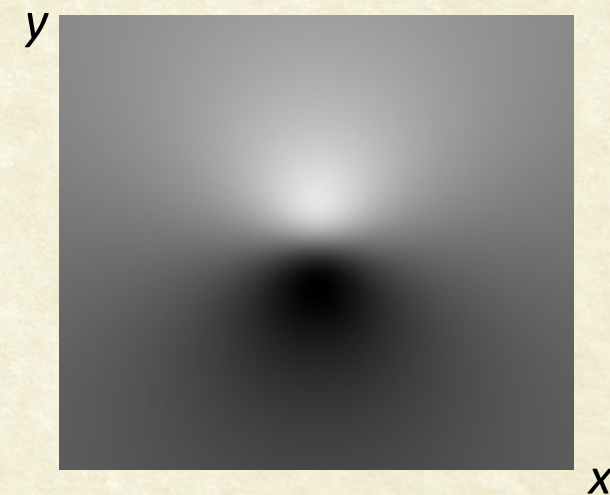
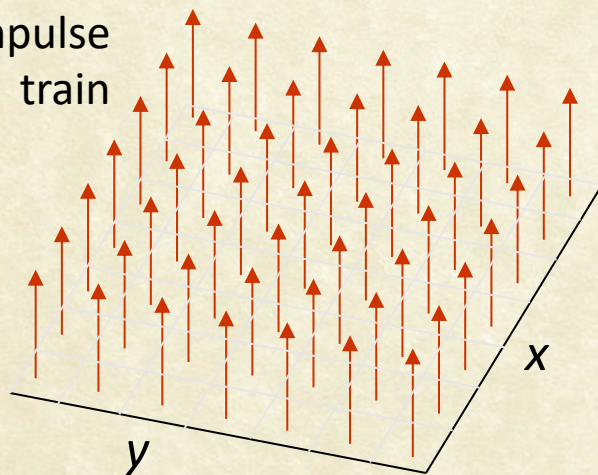


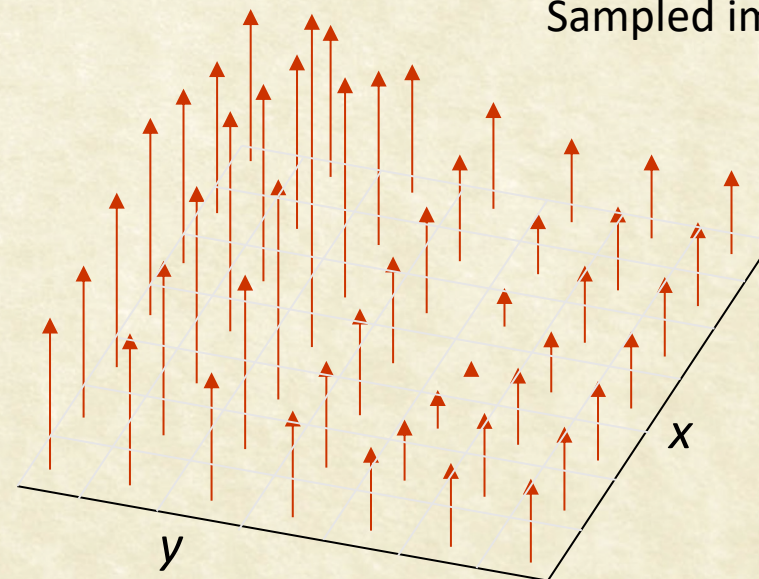
Image formed by the lens on the sensor



2D impulse train

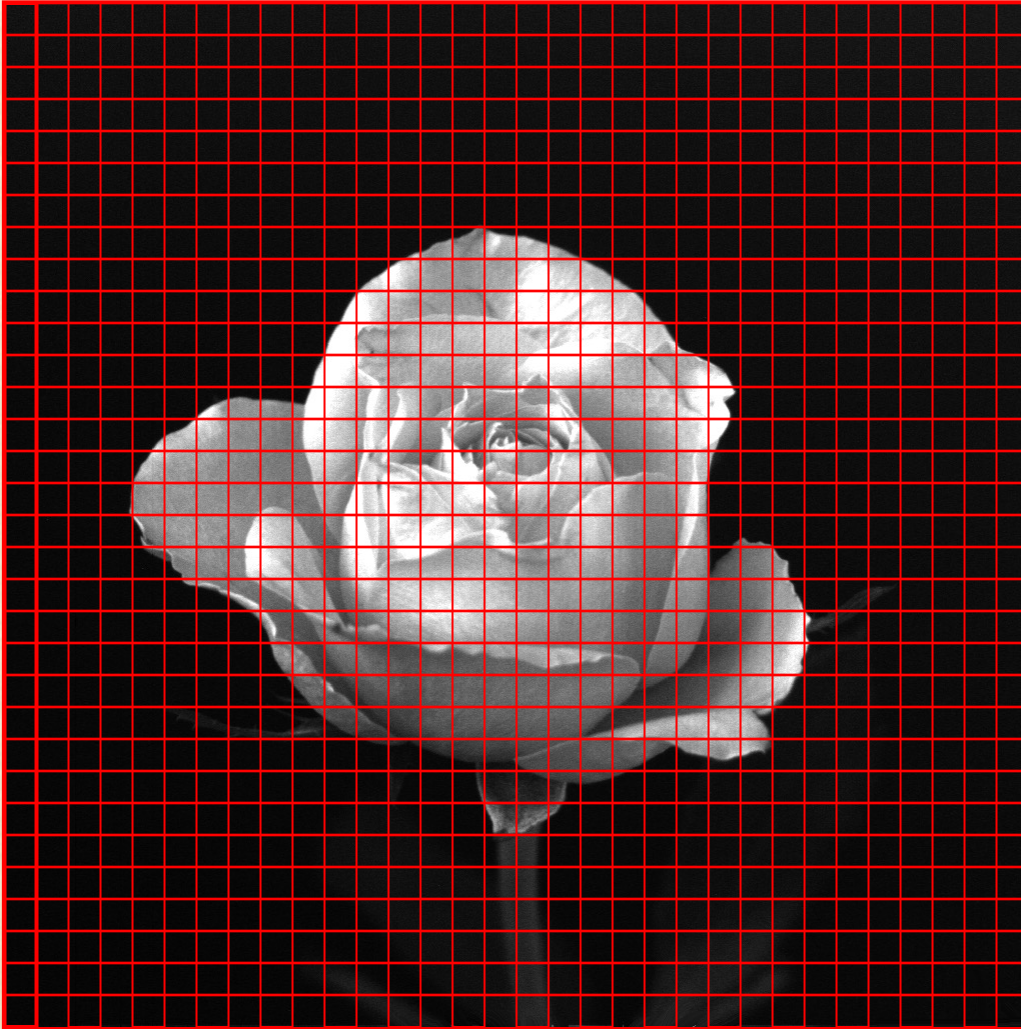


Sampled image





Sampling an Image





Full HD

1920 x 1080

4K

3840 x 2160

5K

5120 x 2160

8K

7680 x 4320



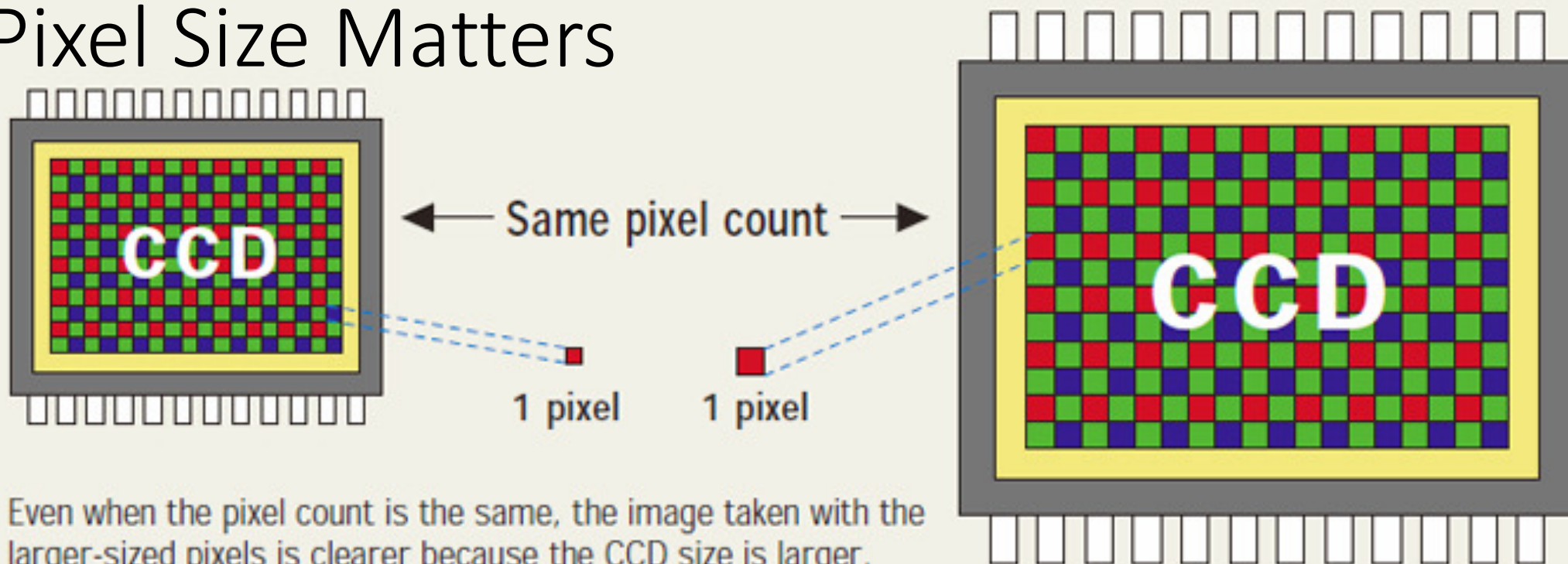


Effect of Resolution





Pixel Size Matters



Even when the pixel count is the same, the image taken with the larger-sized pixels is clearer because the CCD size is larger.





Aspect Ratios

- The ratio of horizontal to vertical image dimensions

Key Terms

Standard Video Qualities: Standard combinations of aspect ratio and video resolution

Video Quality	4:3 Aspect Ratio Resolution	16:9 Aspect Ratio Resolution
360p	480 x 360	640 x 360
480p	640 x 480	854 x 480
720p	Not generally used	1280 x 720
1080p	Not generally used	1920 x 1280

TV ASPECT RATIOS

4:3 (1.33:1)
Standard (SDTV)

16:9 (1.78:1)
Wide Screen (HDTV)

MOVIE THEATER ASPECT RATIOS

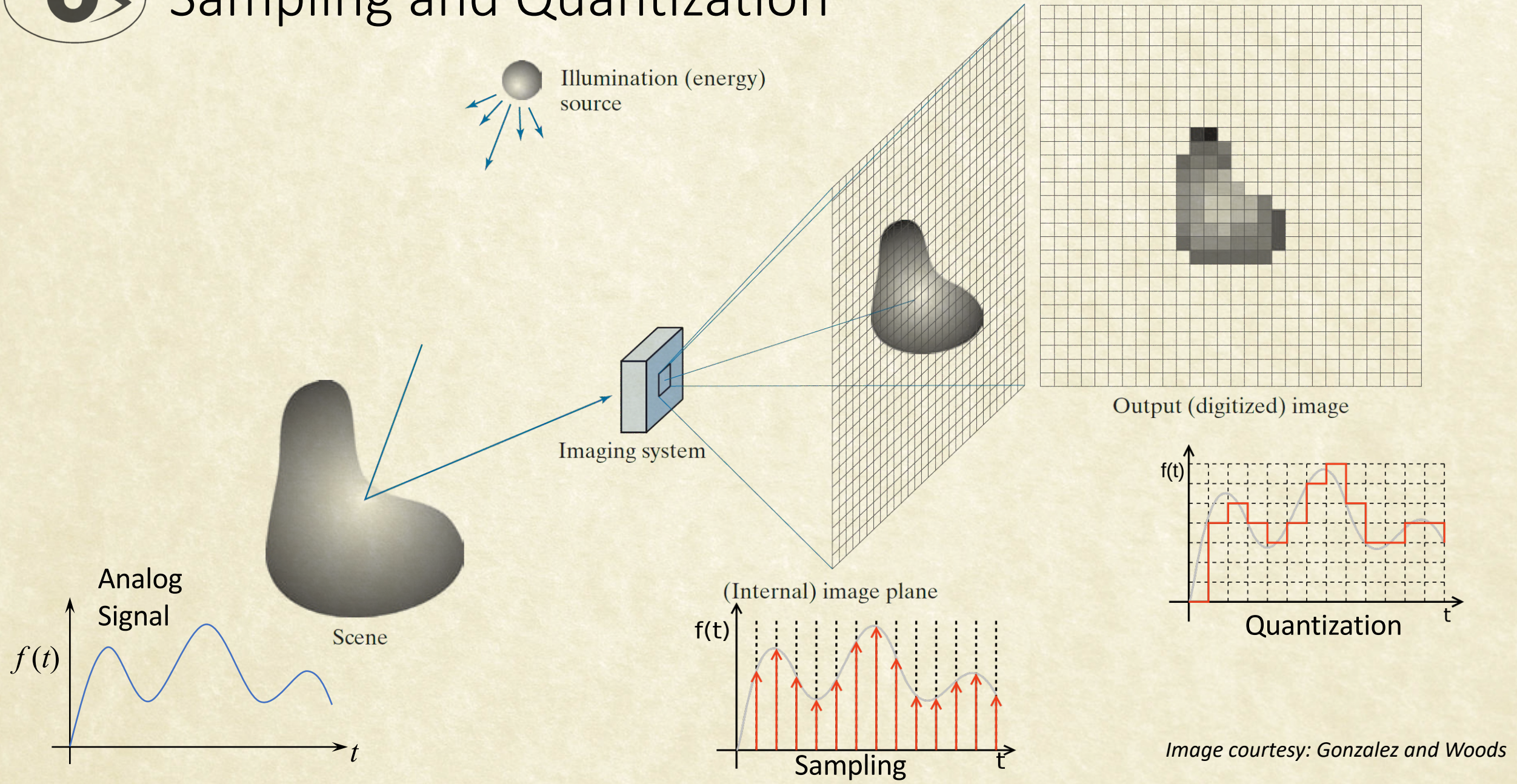
1.37:1
Academy Standard

1.85:1
Academy Flat

2.39:1 (2.35:1 prior to 1970)
Anamorphic Scope
(Panavision/Cinemascope)



Sampling and Quantization

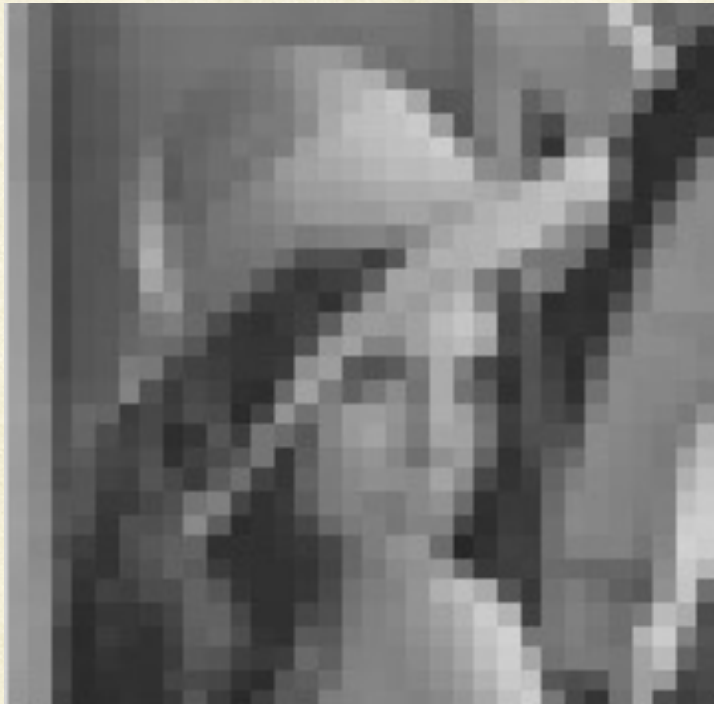




Sampling = Spatial Quantization



256×256



32×32



16×16



Intensity Quantization



8 bits per pixel



4 bits per pixel



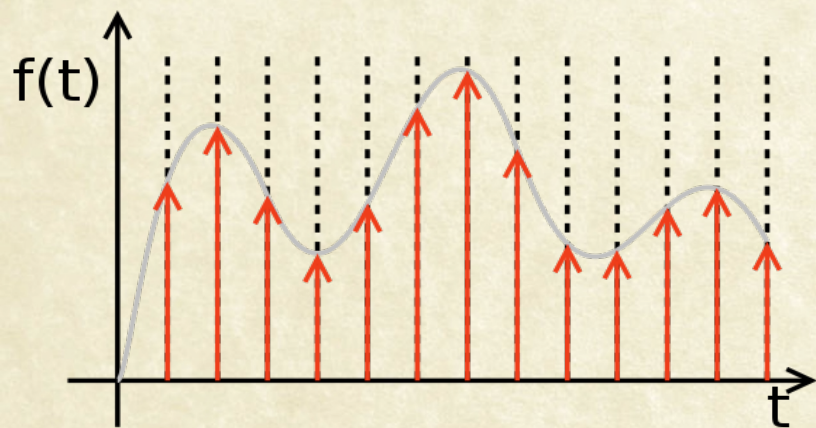
2 bits per pixel



1 bit per pixel



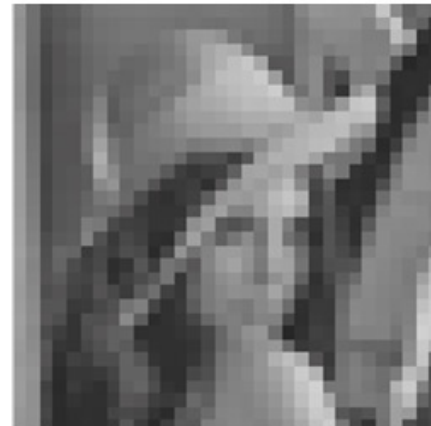
Summary



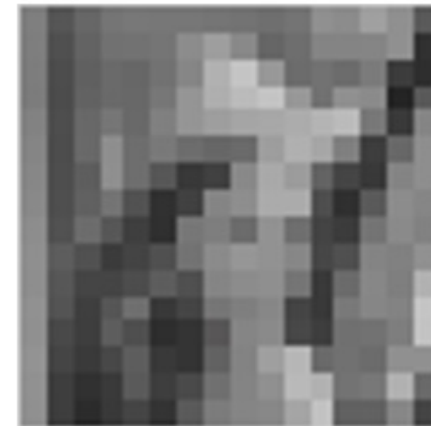
Sampling



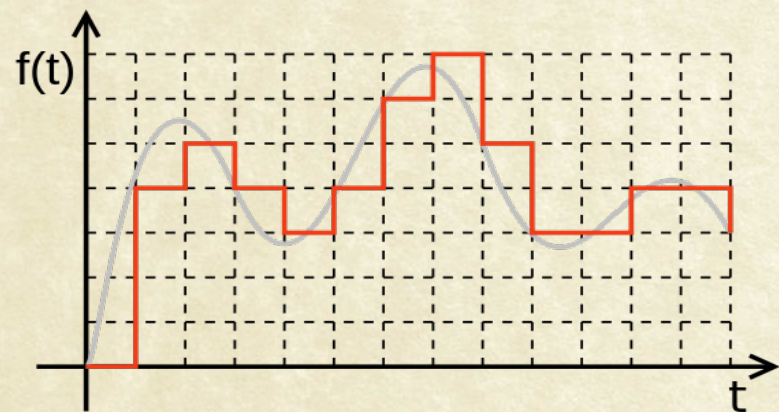
256 × 256



32 × 32



16 × 16



Quantization



8 bits per pixel



4 bits per pixel



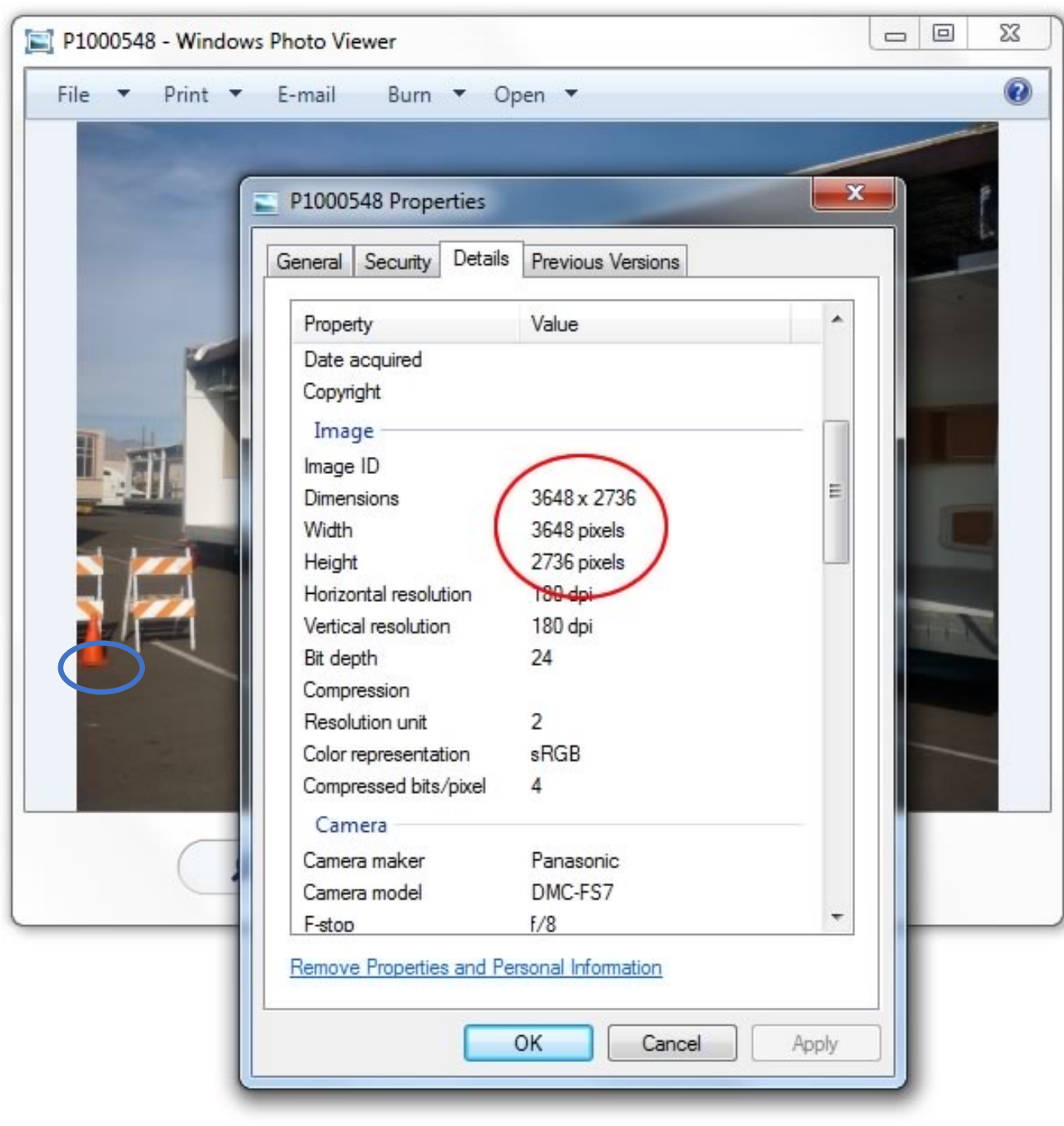
2 bits per pixel



1 bit per pixel



Image Properties





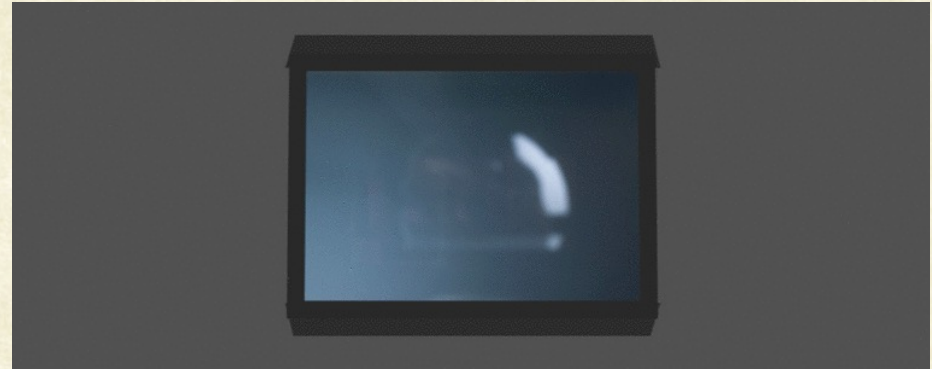
Quantization

- In Hardware (# of voltage levels, # of bits)
- RAW has 24bits or higher capacity (pixels many not have data)
- JPEG is limited to fewer bits per pixel



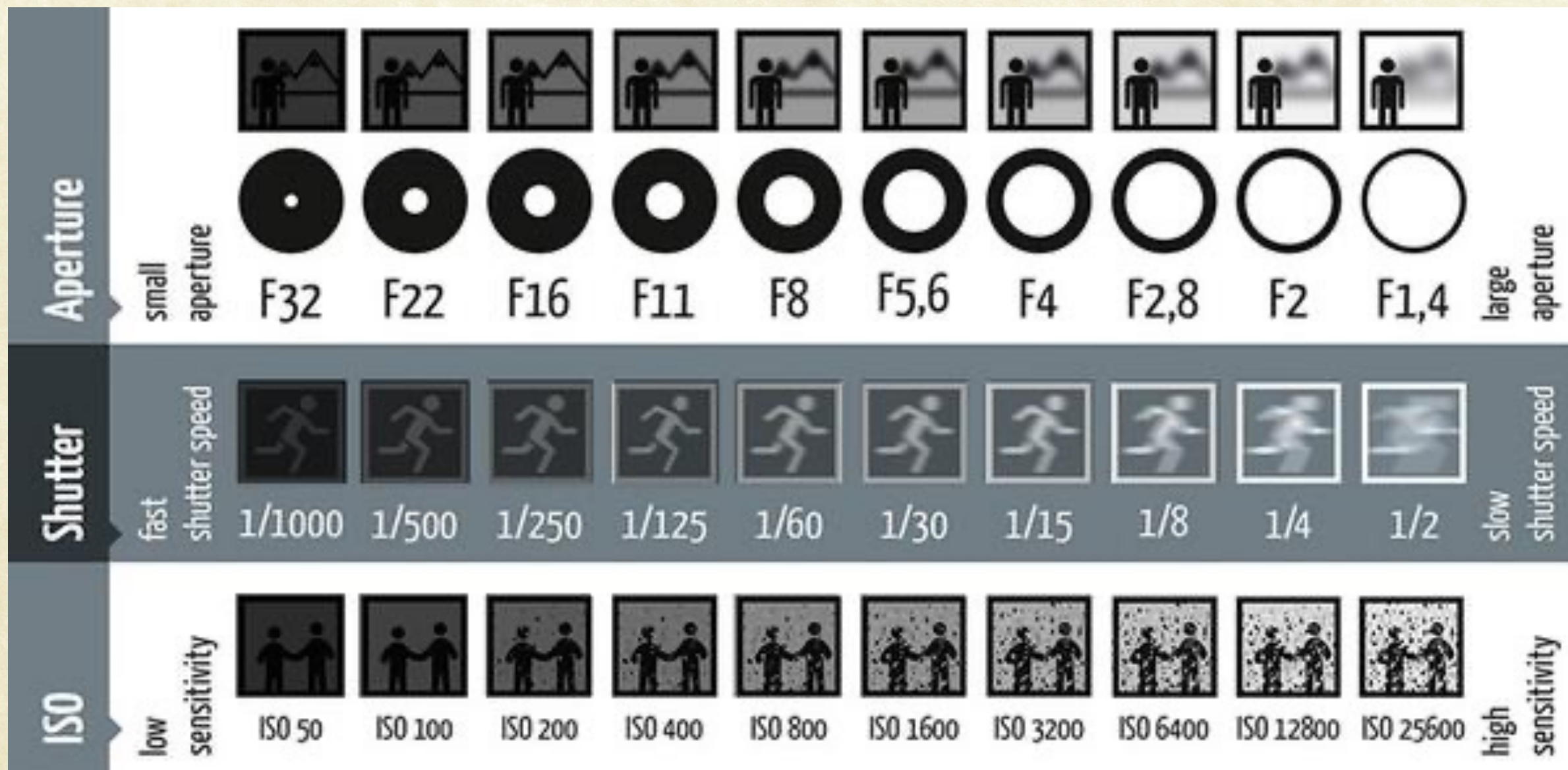


Temporal Sampling



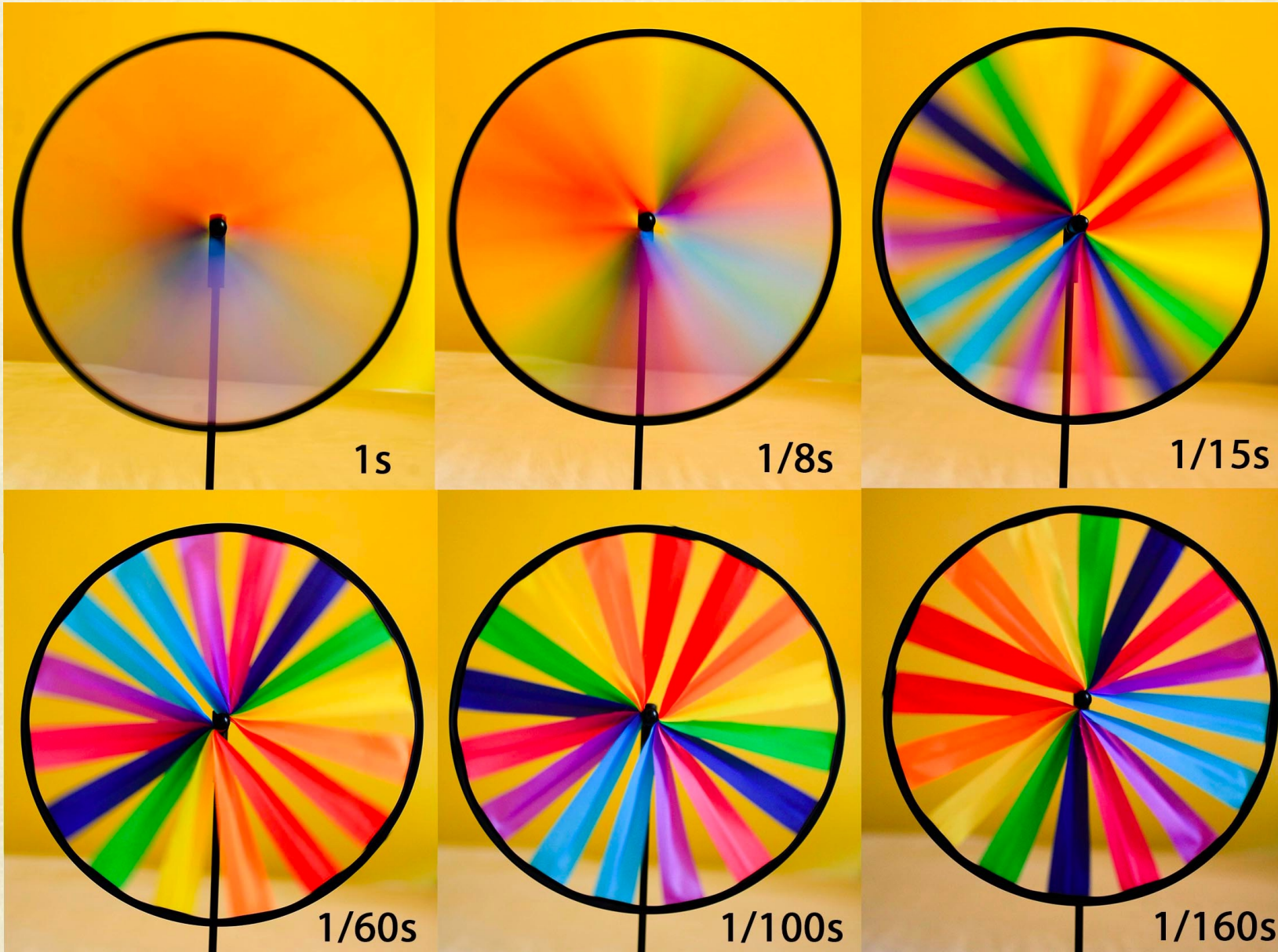


Imaging Controls





Temporal Sampling





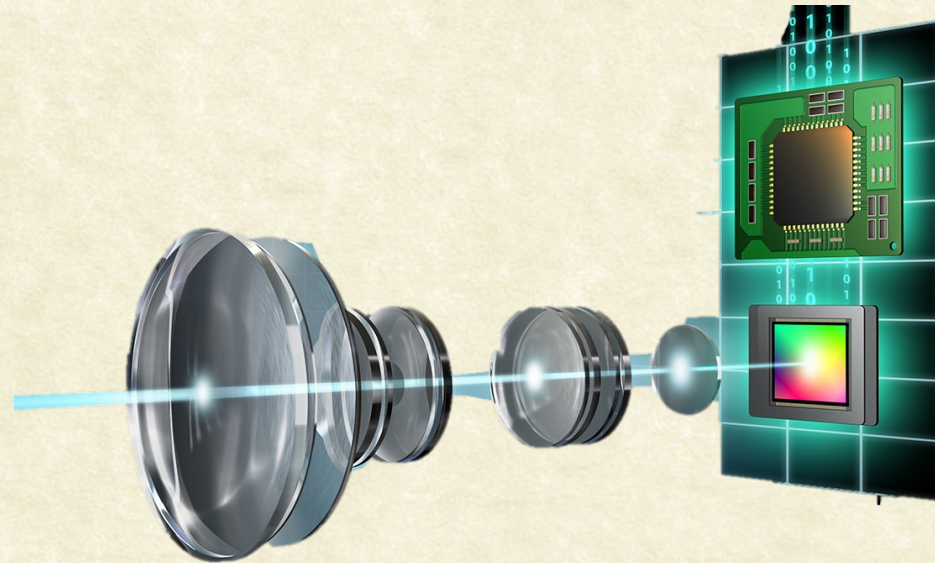
Temporal Sampling





CS7.404: Digital Image Processing

Monsoon 2025: **IV**: Pixel Manipulation



Anoop M. Namboodiri
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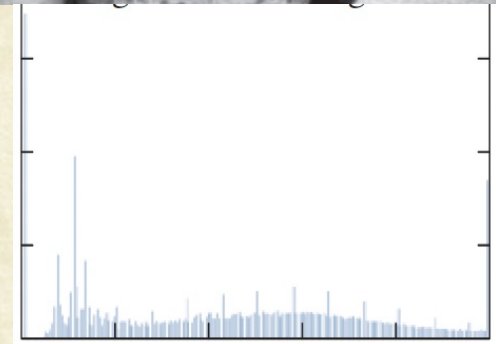
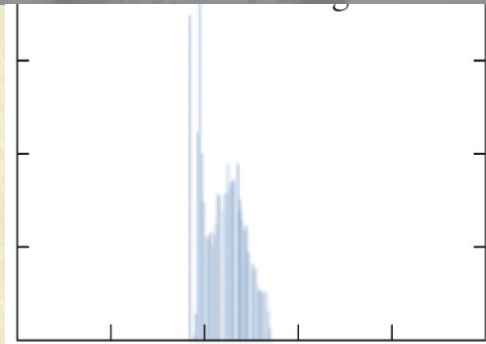
Changing Brightness

$$I'(x, y) = k I(x, y)$$





Contrast Stretching





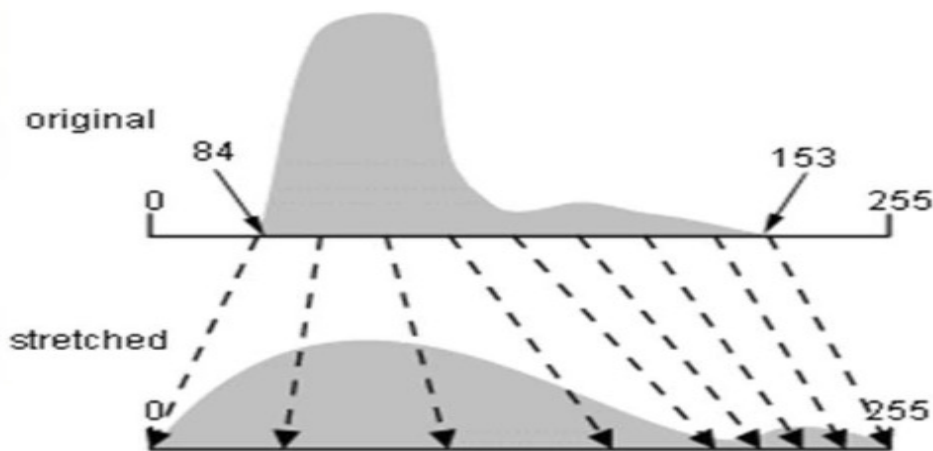
Contrast Stretching



$$f_{\text{ac}}(a) = a_{\text{min}} + (a - a_{\text{low}}) \cdot \frac{a_{\text{max}} - a_{\text{min}}}{a_{\text{high}} - a_{\text{low}}}$$

If $a_{\text{min}} = 0$ and $a_{\text{max}} = 255$

$$f_{\text{ac}}(a) = (a - a_{\text{low}}) \cdot \frac{255}{a_{\text{high}} - a_{\text{low}}}$$



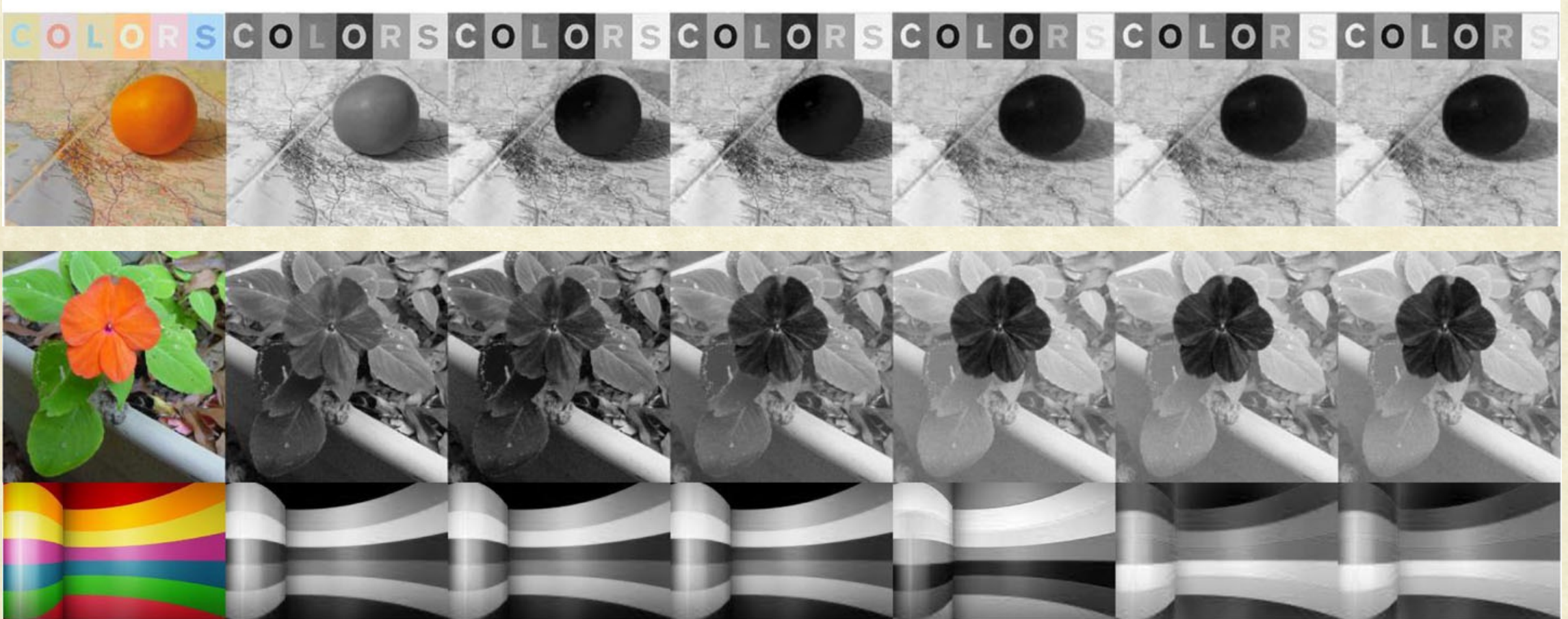
What happens if we have a single pixel with intensity 255 in the original intensity range?

What happens if we have a single pixel with intensity 0 in the original intensity range?



Color to Grayscale

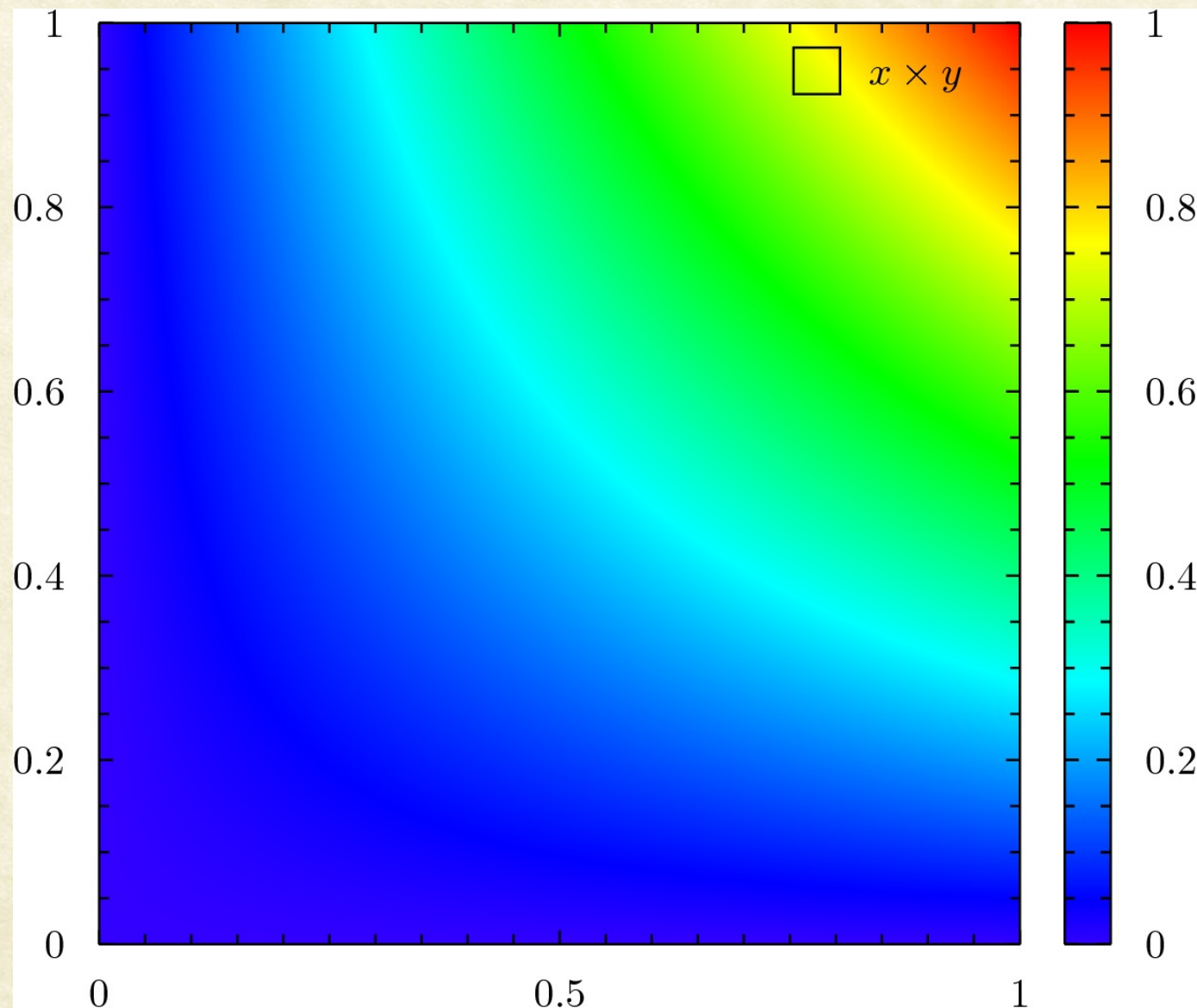
$$I'(x, y) = \alpha I_R(x, y) + \beta I_G(x, y) + \gamma I_B(x, y)$$





Grayscale to Color

$$I'(x, y) = M(I(x, y))$$





Green Screen

- Also known as Chroma Keying
- How to identify background pixels?
- What about boundary pixels?





Image Transition

$$F_t(x, y) = (1 - t) I_1(x, y) + t I_2(x, y)$$

$$t = n/N; \quad n = 0 \dots N$$





Questions?